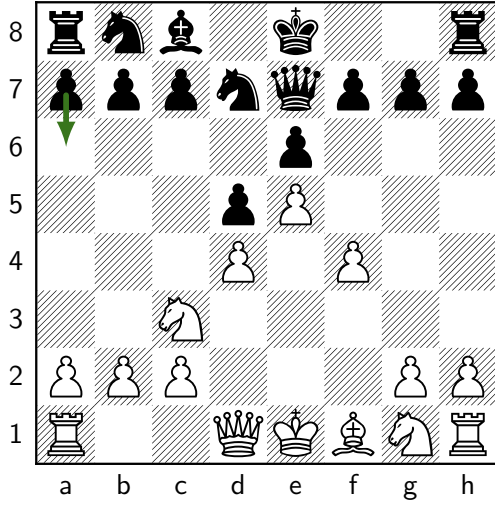


Melanie-XRAY — French, Classical

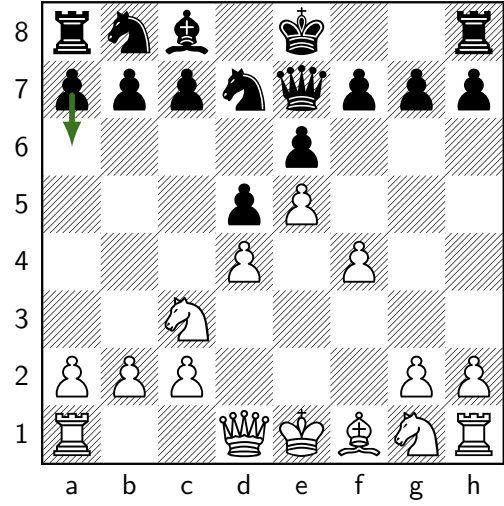
Melanie (Deep_Vision)

Abstract

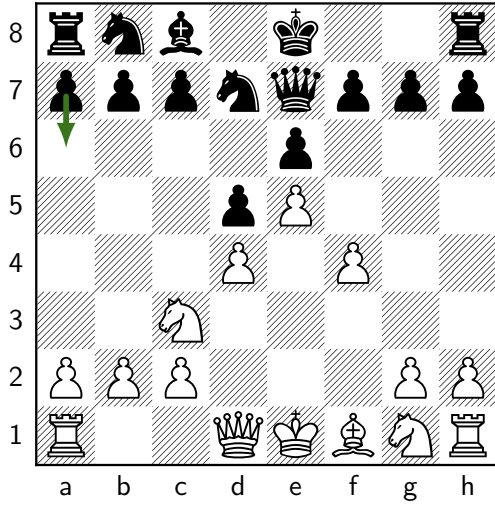
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



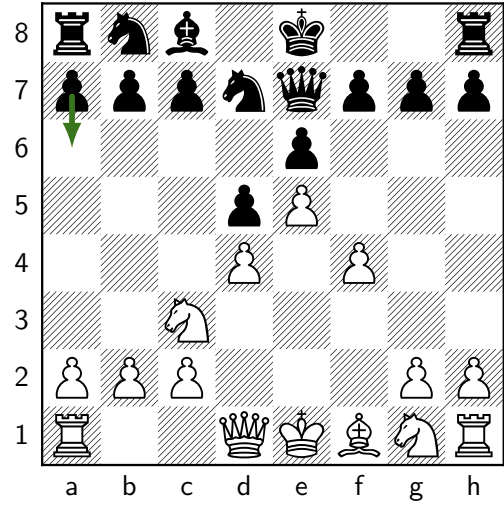
(a)



(b)

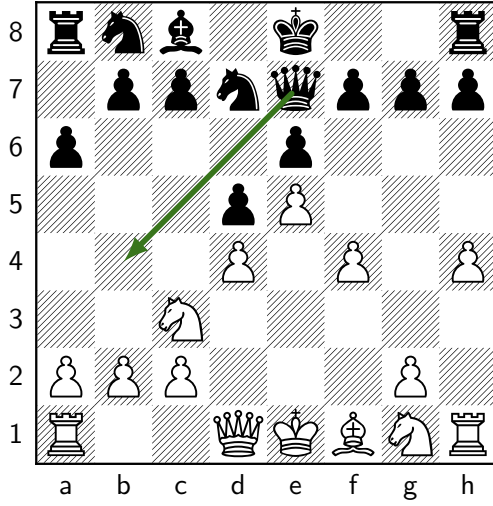


(c)

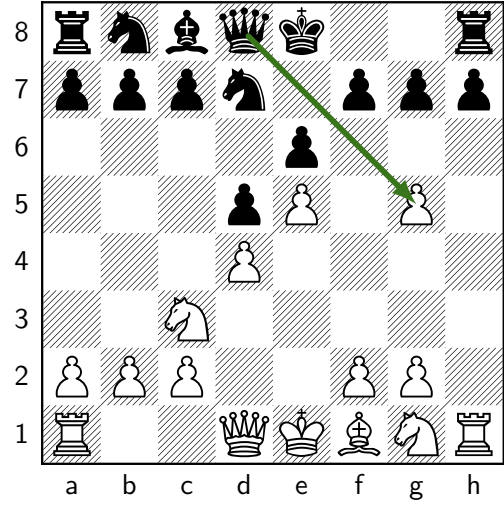


(d)

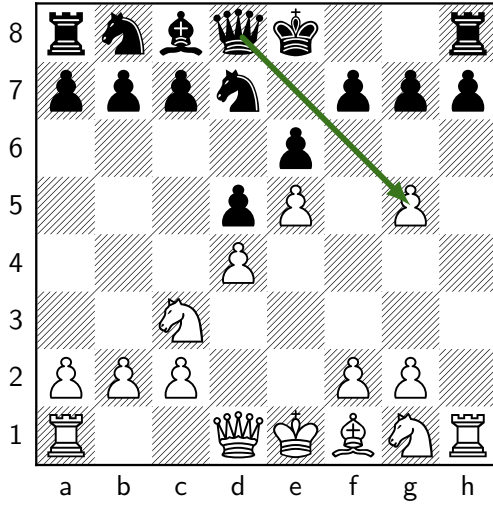
Figure 1: **Move 7, Black: a6 (a7a6).** (a) Our position; Melanie plays a6 (a7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2436` (ply 13) [structural distance $d=0$, QAP obj=64]: the master played a6; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *bf1* as their *ph2*, our *ng1* as their *bf1*; transferred move a7a6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3242` (ply 13) [structural distance $d=0$, QAP obj=64]: the master played a6; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *bf1* as their *ph2*, our *ng1* as their *bf1*; transferred move a7a6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3243` (ply 13) [structural distance $d=0$, QAP obj=64]: the master played a6; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *bf1* as their *ph2*, our *ng1* as their *bf1*; transferred move a7a6 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to a6 (summed QAP objective 422, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



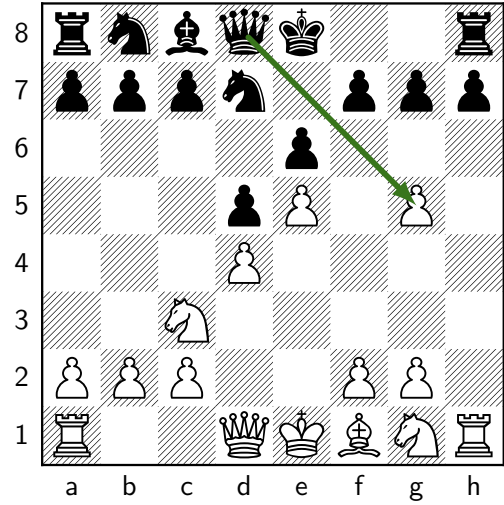
(a)



(b)

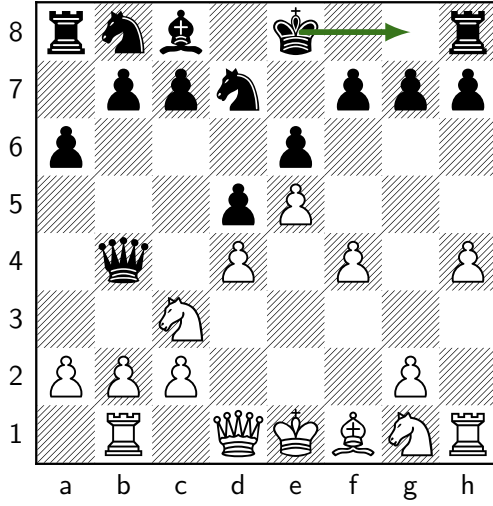


(c)

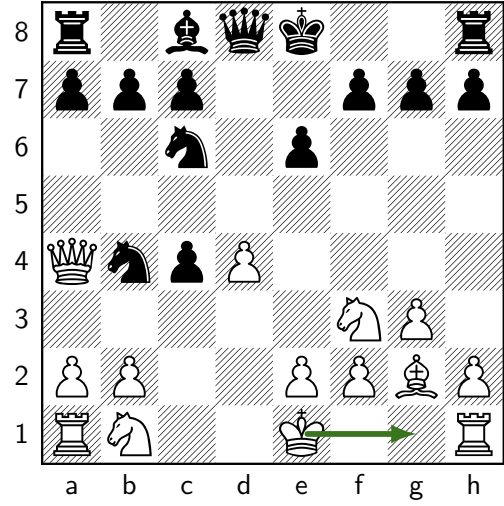


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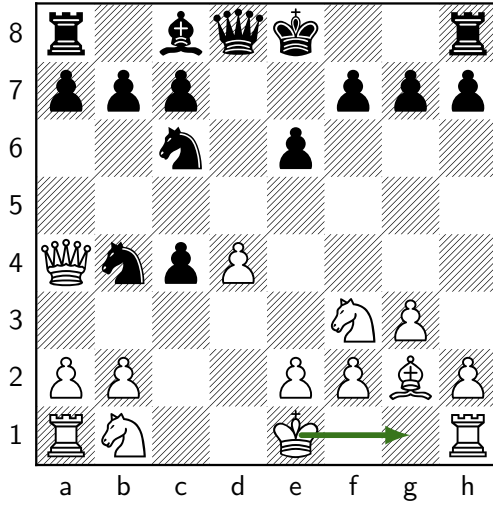
Figure 2: **Move 8, Black: Qb4.** (a) Our position; Melanie plays Qb4 (e7b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#448 (ply 13) [structural distance $d=41$, QAP obj=70]: the master played Qxg5; under μ our Qe2 plays the role of that game’s Qd1; cross-type roles: our Pf7 as their Bc8, our Bc8 as their pf2; transferred move e7b4 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#449 (ply 13) [structural distance $d=41$, QAP obj=70]: the master played Qxg5; under μ our Qe2 plays the role of that game’s Qd1; cross-type roles: our Pf7 as their Bc8, our Bc8 as their pf2; transferred move e7b4 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#1464 (ply 13) [structural distance $d=41$, QAP obj=70]: the master played Qxg5; under μ our Qe2 plays the role of that game’s Qd1; cross-type roles: our Pf7 as their Bc8, our Bc8 as their pf2; transferred move e7b4 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Qb4 (summed QAP objective 448, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



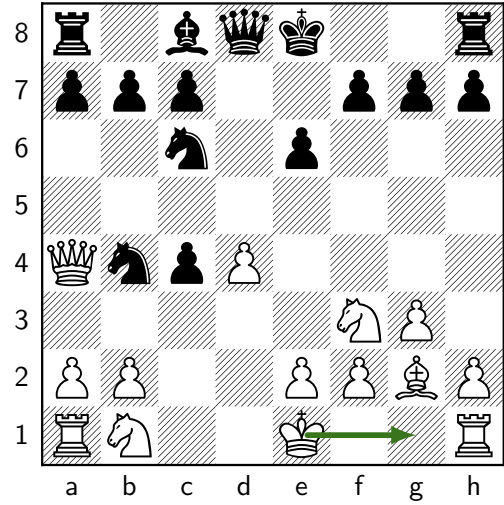
(a)



(b)

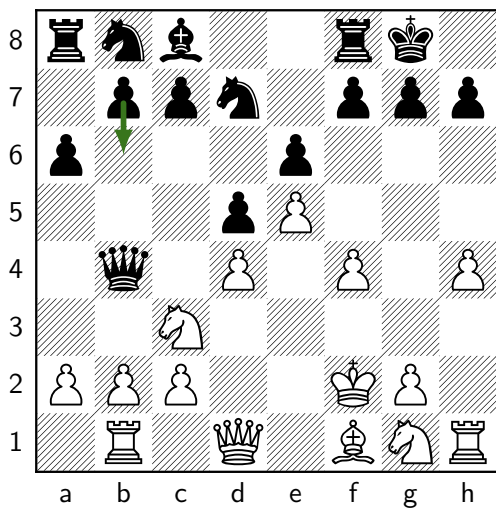


(c)

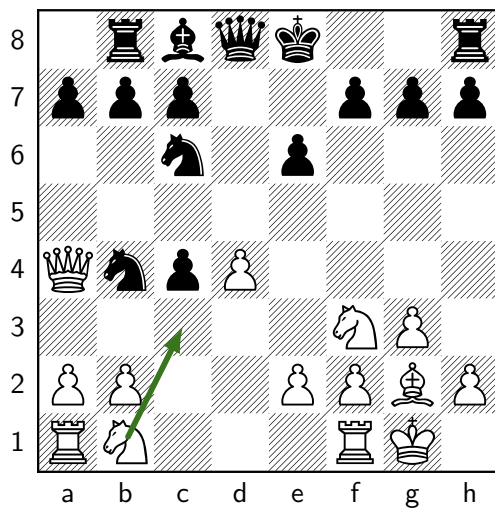


(d)

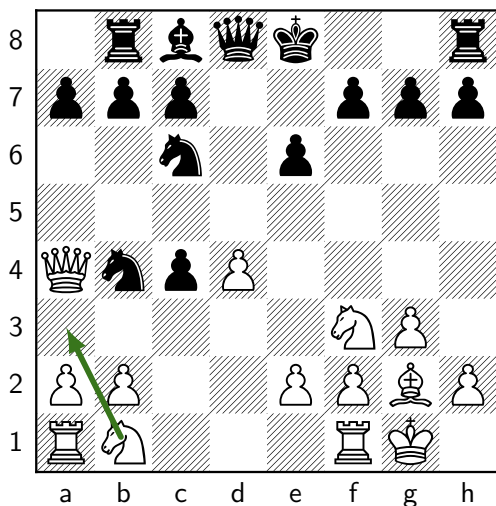
Figure 3: **Move 9, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3568` (ply 16) [structural distance $d=84$, QAP obj=121]: the master played 0-0; under μ our *Ke1* plays the role of that game's *Ke1*; cross-type roles: our *pf4* as their *Ra1*, our *Bc8* as their *Rh1*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2311` (ply 16) [structural distance $d=84$, QAP obj=121]: the master played 0-0; under μ our *Ke1* plays the role of that game's *Ke1*; cross-type roles: our *pf4* as their *Ra1*, our *Bc8* as their *Rh1*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1576` (ply 16) [structural distance $d=84$, QAP obj=121]: the master played 0-0; under μ our *Ke1* plays the role of that game's *Ke1*; cross-type roles: our *pf4* as their *Ra1*, our *Bc8* as their *Rh1*; transferred move e8g8 (**endorses this move**). Synthesis: 24 retrieved analogs transferred to 0-0 (summed QAP objective 1578, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



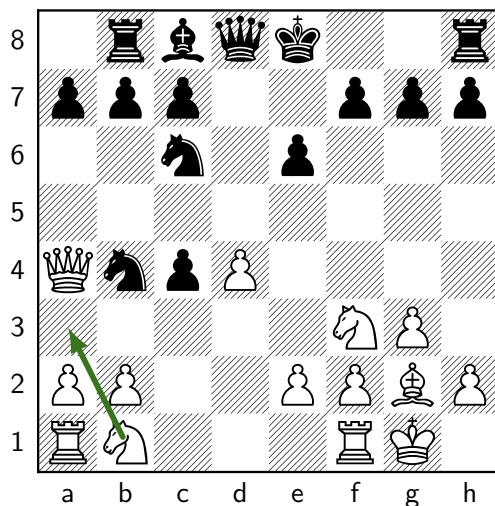
(a)



(b)

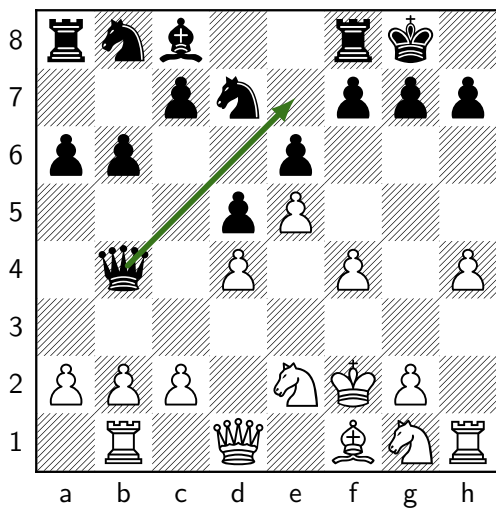


(c)

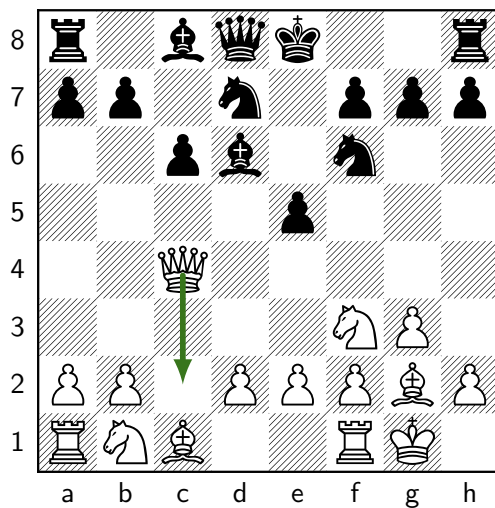


(d)

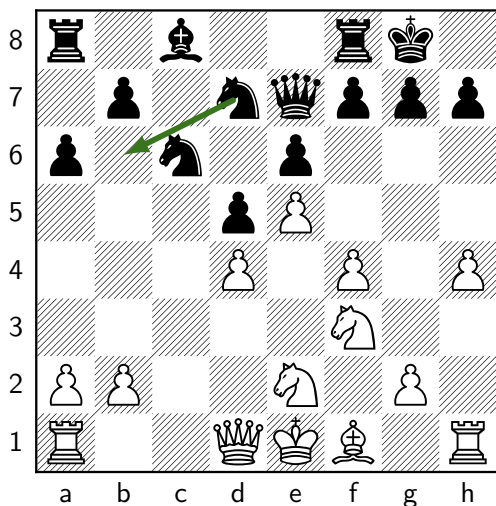
Figure 4: **Move 10, Black: b6.** (a) Our position; Melanie plays b6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3568` (ply 18) [structural distance $d=91$, QAP obj=76]: the master played Nc3; under μ our *Pb2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Nd7* as their *Bg2*, our *pf4* as their *Nf3*; transferred move b7b6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2311` (ply 18) [structural distance $d=91$, QAP obj=76]: the master played Na3; under μ our *Pb2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Nd7* as their *Bg2*, our *pf4* as their *Nf3*; transferred move b7b6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1576` (ply 18) [structural distance $d=91$, QAP obj=76]: the master played Na3; under μ our *Pb2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Nd7* as their *Bg2*, our *pf4* as their *Nf3*; transferred move b7b6 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to b6 (summed QAP objective 630, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



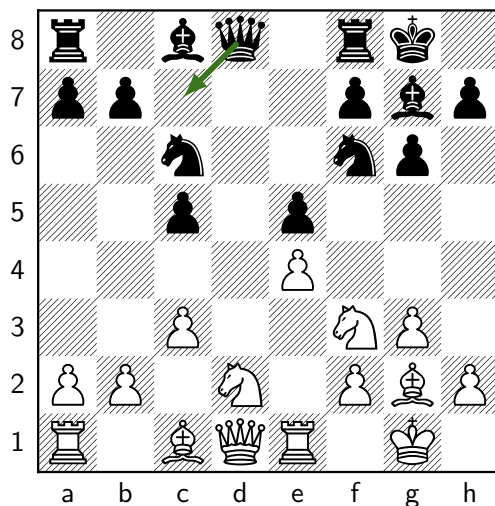
(a)



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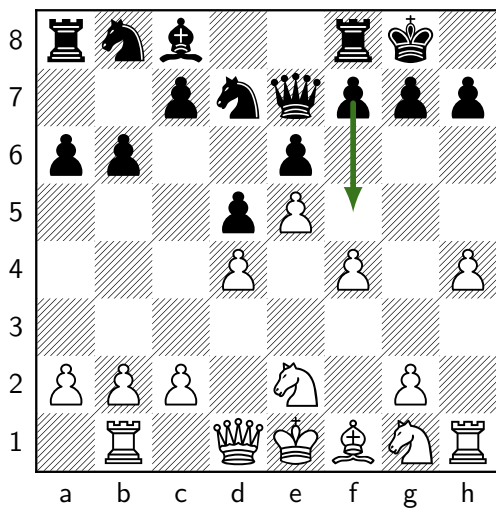


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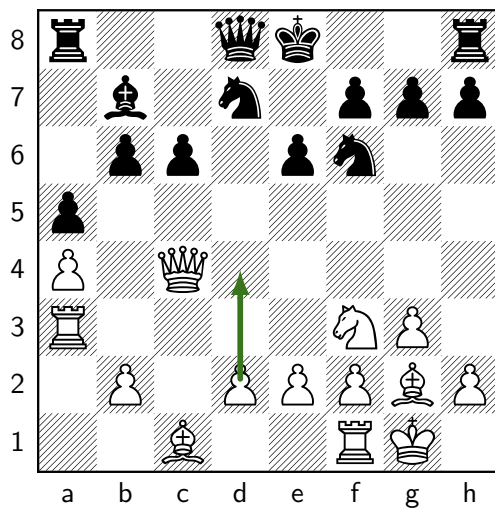


(d)

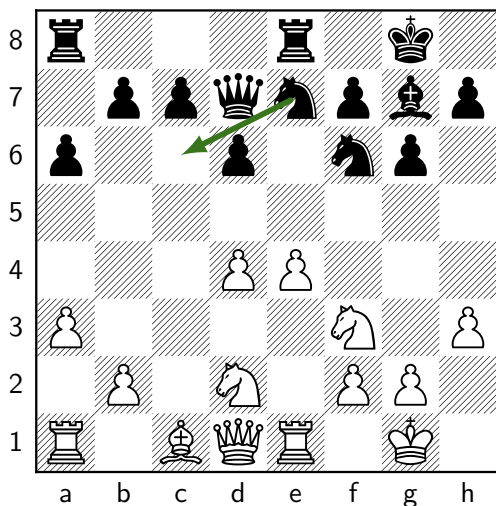
Figure 5: **Move 11, Black: Qe7.** (a) Our position; Melanie plays Qe7 (b4e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#211` (ply 14) [structural distance $d=105$, QAP obj=86]: the master played Qc2; under μ our Qb5 plays the role of that game’s Qc4; cross-type roles: our Pd5 as their Nb1, our Rf8 as their Bc1; transferred move b4e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2436` (ply 23) [structural distance $d=107$, QAP obj=79]: the master played Nb6; under μ our Qb5 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Nd7 as their Qe7, our Nb8 as their Pa6; transferred move b4e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2405` (ply 19) [structural distance $d=98$, QAP obj=79]: the master played Qc7; under μ our Qb5 plays the role of that game’s Qd1; cross-type roles: our Pe6 as their Bc8, our Pb6 as their Rf8; transferred move b4e7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qe7 (summed QAP objective 243, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



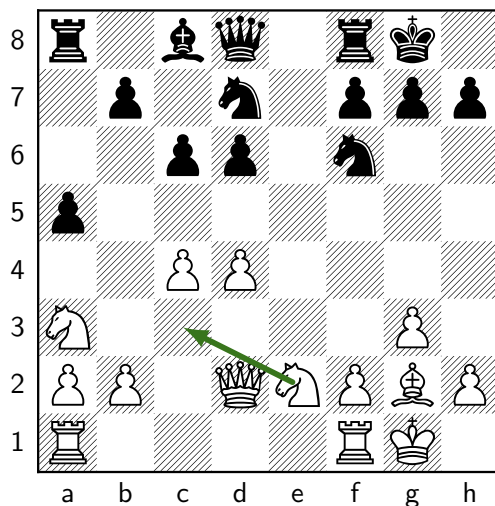
(a)



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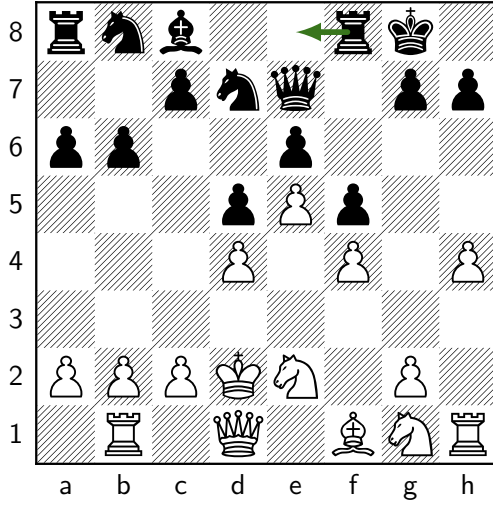


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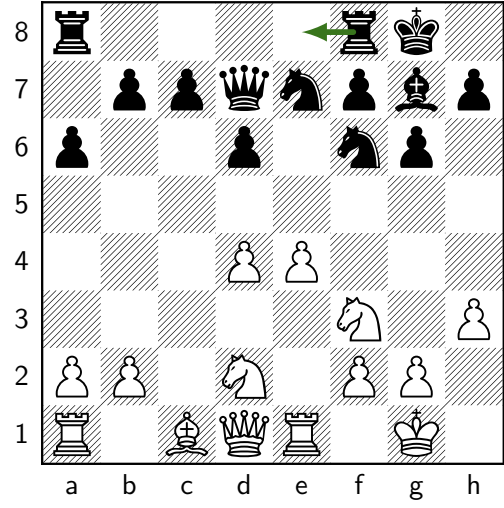


(d)

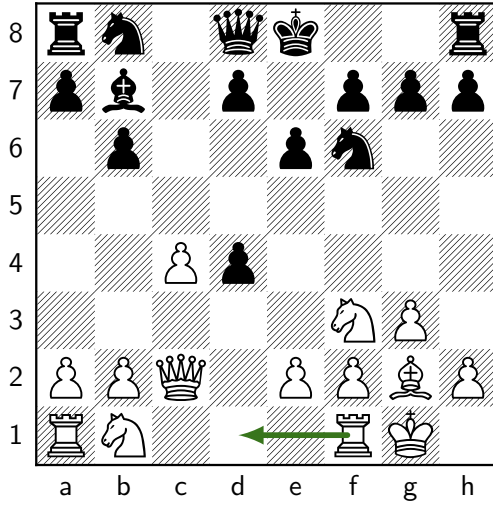
Figure 6: **Move 12, Black: f5.** (a) Our position; Melanie plays f5 (f7f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3243` (ply 20) [structural distance $d=86$, QAP obj=39]: the master played d4; under μ our $Pf2$ plays the role of that game’s $Pd2$; cross-type roles: our $Nb8$ as their $Bc1$, our $Ra8$ as their $Pb2$; transferred move f7f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3648` (ply 27) [structural distance $d=78$, QAP obj=38]: the master played Nc6; under μ our $Pf2$ plays the role of that game’s $Ne2$ (a **cross-type** correspondence); cross-type roles: our $Nb8$ as their $Pb7$, our $Bc8$ as their $Pf7$; transferred move f7f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#32` (ply 22) [structural distance $d=85$, QAP obj=33]: the master played Nc3; under μ our $Pf2$ plays the role of that game’s $Ne2$ (a **cross-type** correspondence); cross-type roles: our $Nb8$ as their $Pa2$, our $Bc8$ as their $Pb2$; transferred move f7f5 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to f5 (summed QAP objective 308, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



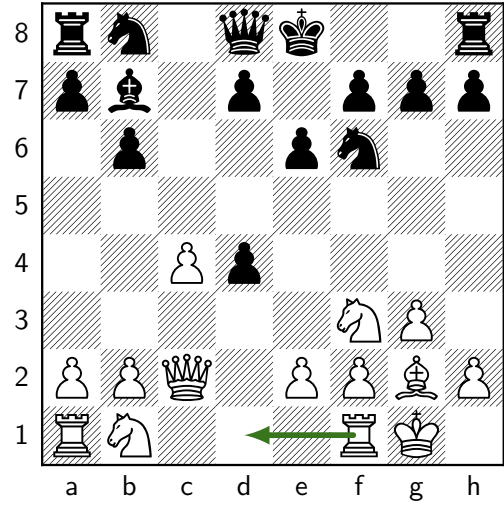
(a)



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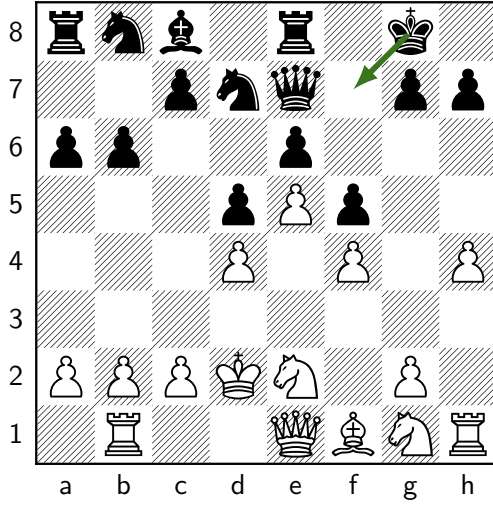


(c)

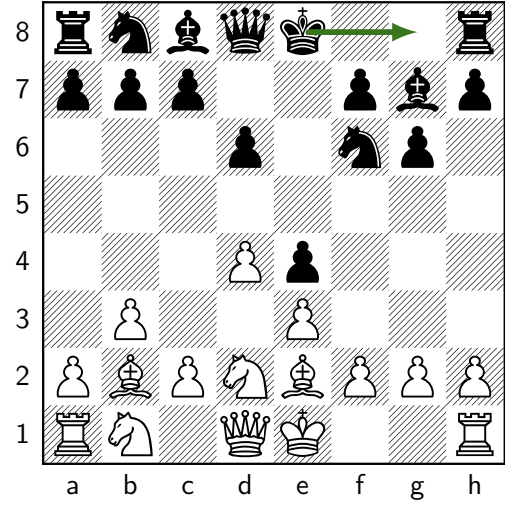


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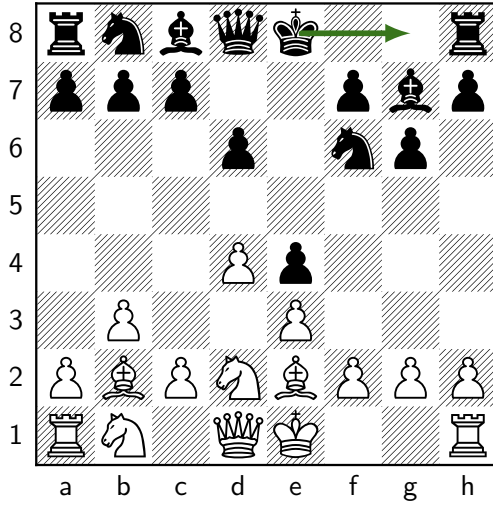
Figure 7: **Move 13, Black: Re8.** (a) Our position; Melanie plays **Re8 (f8e8)**. Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3648` (ply 25) [structural distance $d=79$, QAP obj=37]: the master played **Rfe8**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Pa6* as their *Ra8*, our *Pe6* as their *Ne7*; transferred move **f8e8 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#2015` (ply 18) [structural distance $d=90$, QAP obj=25]: the master played **Rd1**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Pg7* as their *Ra1*, our *Bc8* as their *Pa2*; transferred move **f8e8 (endorses this move)**. (d) Retrieved from `ccr1_top_p0000#2017` (ply 18) [structural distance $d=90$, QAP obj=25]: the master played **Rd1**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Pg7* as their *Ra1*, our *Bc8* as their *Pa2*; transferred move **f8e8 (endorses this move)**. Synthesis: 9 retrieved analogs transferred to **Re8** (summed QAP objective 204, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



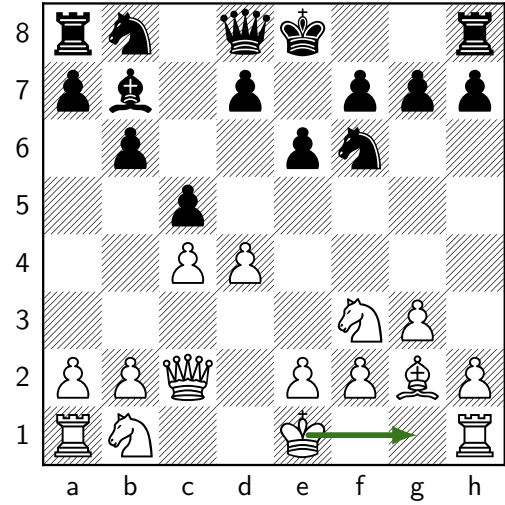
(a)



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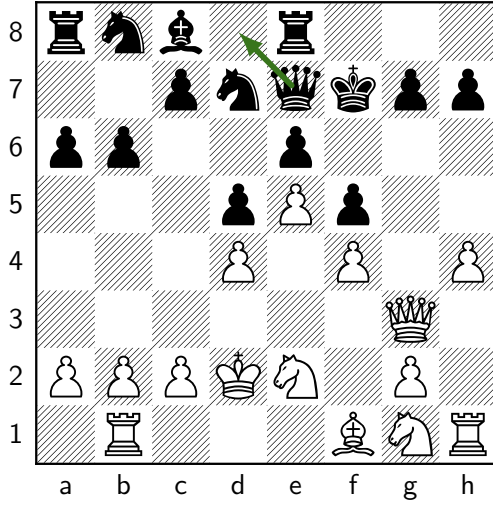


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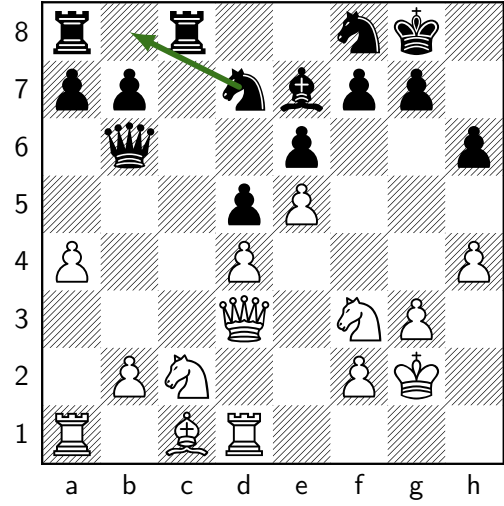


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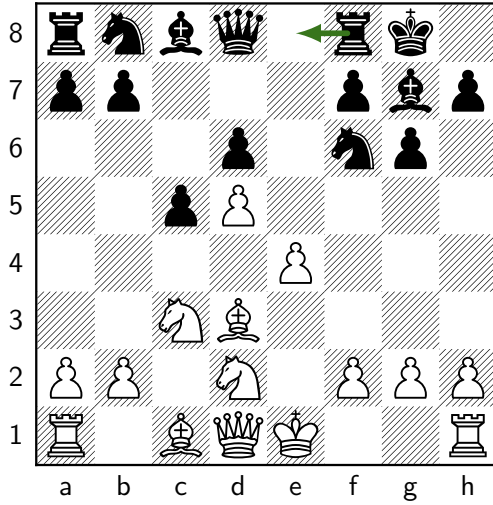
Figure 8: **Move 14, Black: Kf7.** (a) Our position; Melanie plays Kf7 (g8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1948` (ply 13) [structural distance $d=88$, QAP obj=21]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qe7* as their *Bc8*, our *Pg7* as their *Qd8*; transferred move g8f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1949` (ply 13) [structural distance $d=88$, QAP obj=21]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qe7* as their *Bc8*, our *Pg7* as their *Qd8*; transferred move g8f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2015` (ply 16) [structural distance $d=90$, QAP obj=19]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Pc7* as their *Rh1*, our *Re8* as their *Pa2*; transferred move g8f7 (**endorses this move**). Synthesis: 15 retrieved analogs transferred to Kf7 (summed QAP objective 282, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



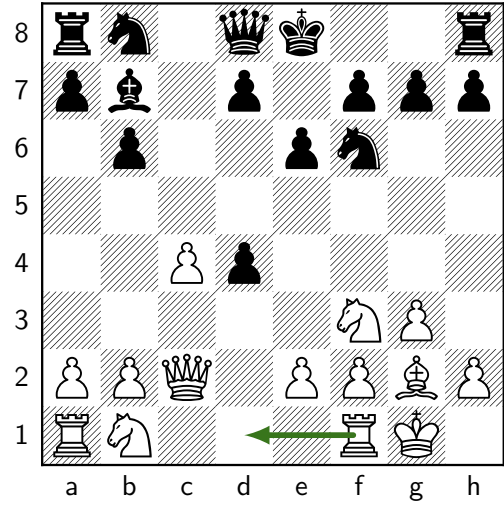
(a)



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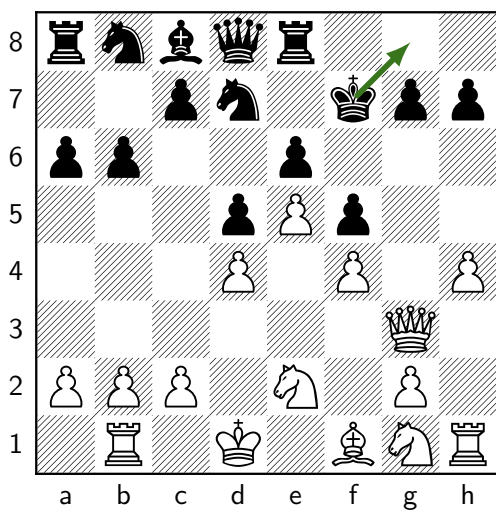


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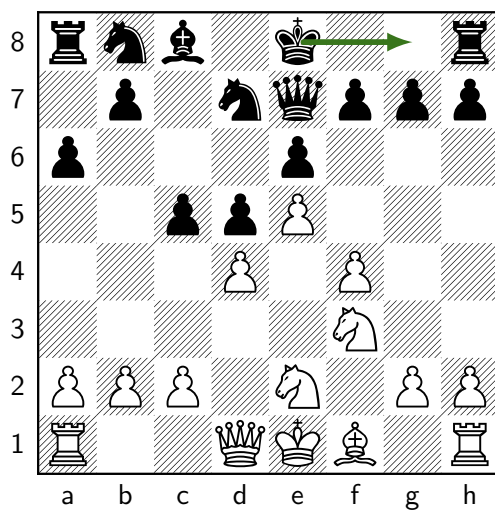


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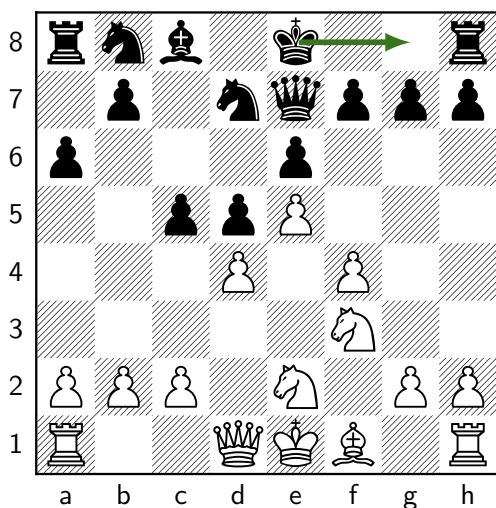
Figure 9: **Move 15, Black: Qd8.** (a) Our position; Melanie plays Qd8 (e7d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1897` (ply 35) [structural distance $d=92$, QAP obj=62]: the master played Nb8; under μ our Qe2 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Pe6 as their Ra8, our Nd7 as their Rc8; transferred move e7d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3645` (ply 17) [structural distance $d=98$, QAP obj=37]: the master played Re8; under μ our Qe2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Pe6 as their Nb8, our Pc7 as their Bc8; transferred move e7d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2015` (ply 18) [structural distance $d=99$, QAP obj=22]: the master played Rd1; under μ our Qe2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our pf4 as their Bg2, our pe5 as their Nf3; transferred move e7d8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd8 (summed QAP objective 142, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



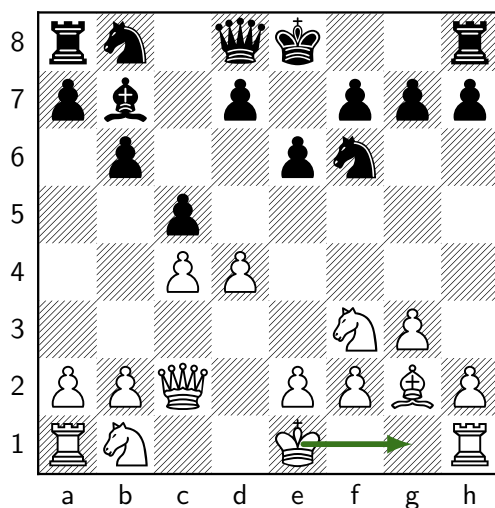
(a)



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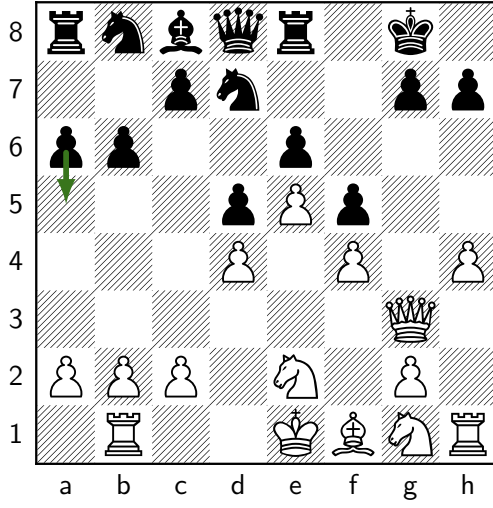


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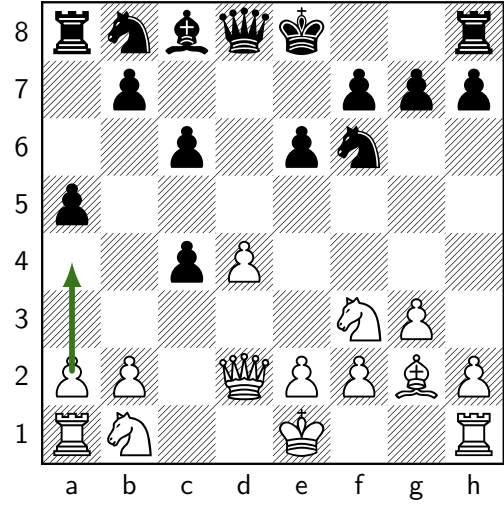


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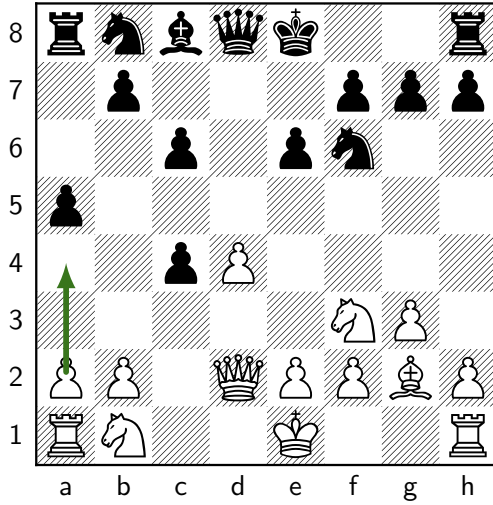
Figure 10: **Move 16, Black: Kg8**. (a) Our position; Melanie plays Kg8 (f7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2436` (ply 17) [structural distance $d=73$, QAP obj=18]: the master played 0-0; under μ our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *pg2* as their *Ra8*, our *rh1* as their *Nb8*; transferred move f7g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3243` (ply 17) [structural distance $d=73$, QAP obj=18]: the master played 0-0; under μ our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *pg2* as their *Ra8*, our *rh1* as their *Nb8*; transferred move f7g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2015` (ply 16) [structural distance $d=88$, QAP obj=18]: the master played 0-0; under μ our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *ne2* as their *Ra1*, our *Pe6* as their *Rh1*; transferred move f7g8 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Kg8 (summed QAP objective 288, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



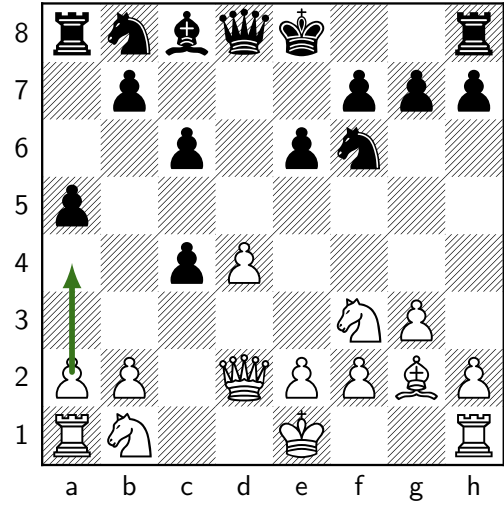
(a)



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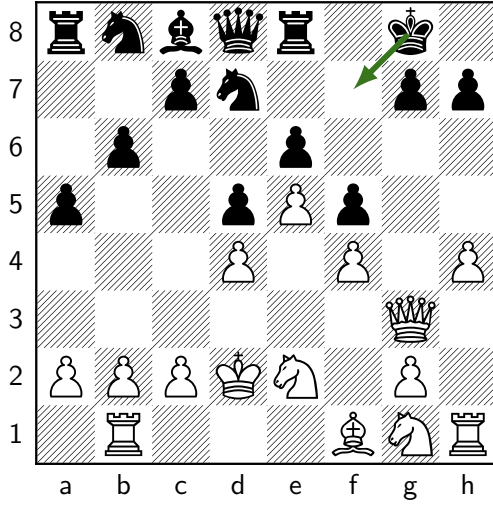


(c)

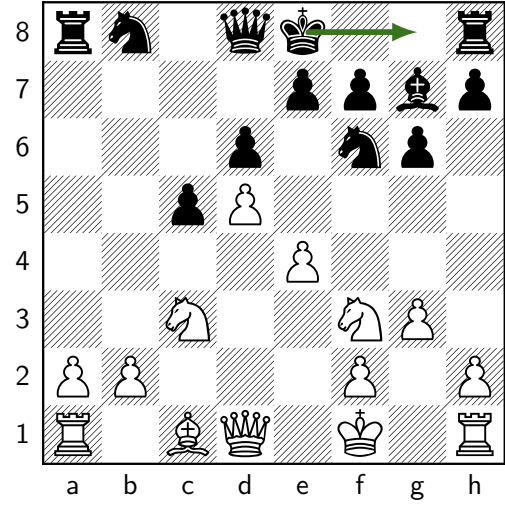


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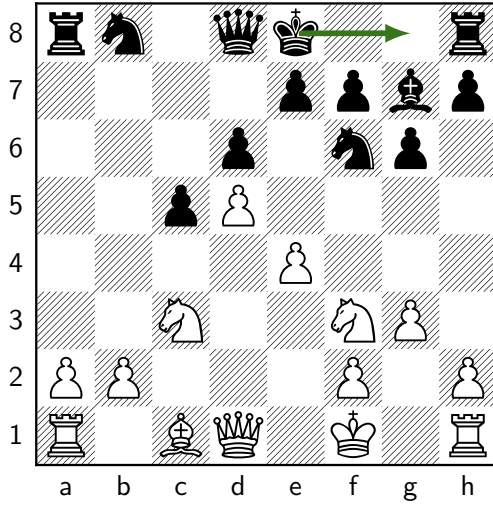
Figure 11: **Move 17, Black: a5.** (a) Our position; Melanie plays a5 (a6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3544` (ply 16) [structural distance $d=82$, QAP obj=50]: the master played a4; under μ our $Pa3$ plays the role of that game's $Pa2$; cross-type roles: our $Qd8$ as their $Ra1$, our $Bc8$ as their $Nb1$; transferred move a6a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#956` (ply 16) [structural distance $d=82$, QAP obj=50]: the master played a4; under μ our $Pa3$ plays the role of that game's $Pa2$; cross-type roles: our $Qd8$ as their $Ra1$, our $Bc8$ as their $Nb1$; transferred move a6a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#957` (ply 16) [structural distance $d=82$, QAP obj=50]: the master played a4; under μ our $Pa3$ plays the role of that game's $Pa2$; cross-type roles: our $Qd8$ as their $Ra1$, our $Bc8$ as their $Nb1$; transferred move a6a5 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to a5 (summed QAP objective 438, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



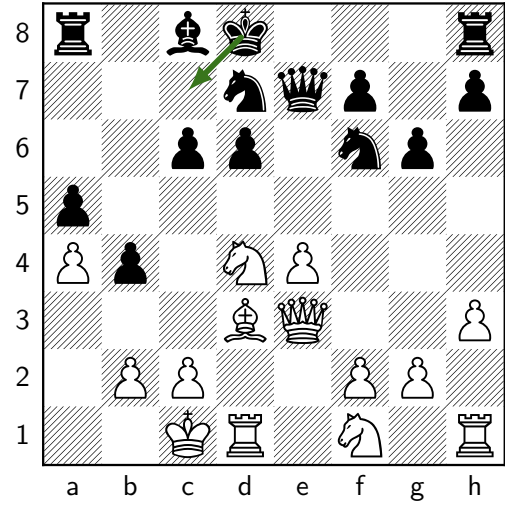
(a)



(b)

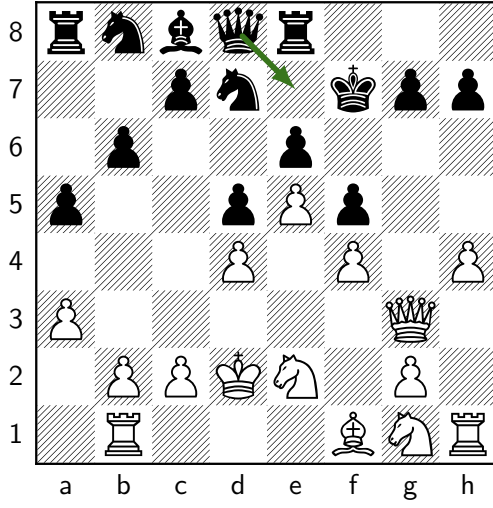


(c)

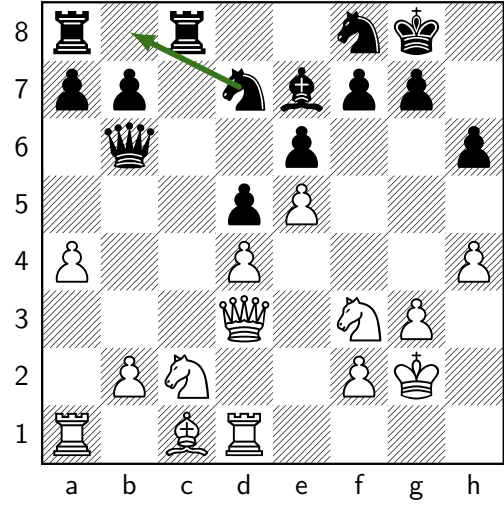


(d)

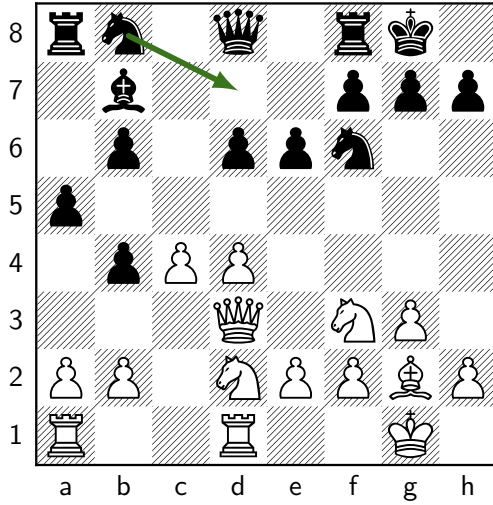
Figure 12: **Move 18, Black: Kf7 (g8f7).** (a) Our position; Melanie plays Kf7 (g8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#802` (ply 19) [structural distance $d=108$, QAP obj=62]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *qg3* as their *Ra8*, our *ph4* as their *Nb8*; transferred move g8f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1851` (ply 19) [structural distance $d=108$, QAP obj=62]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *qg3* as their *Ra8*, our *ph4* as their *Nb8*; transferred move g8f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2706` (ply 33) [structural distance $d=107$, QAP obj=56]: the master played Kc7; under μ our *Kg1* plays the role of that game’s *Kd1*; cross-type roles: our *Qd8* as their *Ra8*, our *pa2* as their *Rh8*; transferred move g8f7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kf7 (summed QAP objective 262, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



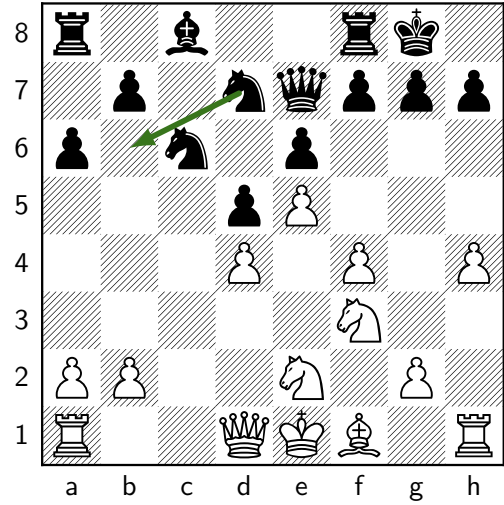
(a)



(b)

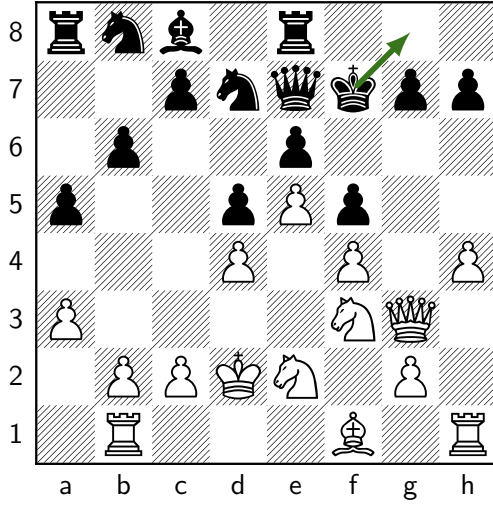


(c)

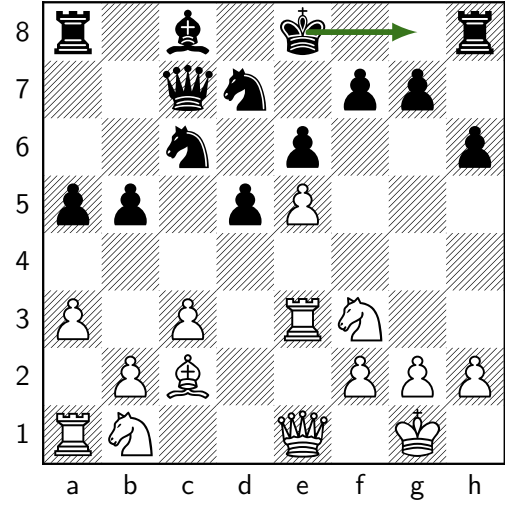


(d)

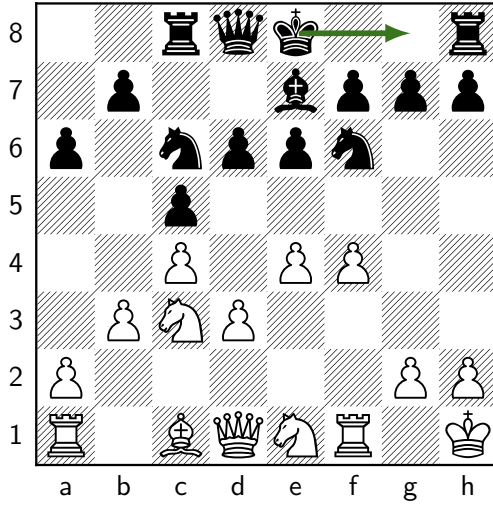
Figure 13: **Move 19, Black: Qe7.** (a) Our position; Melanie plays Qe7 (d8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1897` (ply 35) [structural distance $d=98$, QAP obj=61]: the master played Nb8; under μ our Qd1 plays the role of that game's Nd2 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ra8, our Nd7 as their Rc8; transferred move d8e7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1980` (ply 21) [structural distance $d=97$, QAP obj=21]: the master played Nbd7; under μ our Qd1 plays the role of that game's Nb1 (a **cross-type** correspondence); cross-type roles: our pd4 as their Ra8, our bf1 as their Qd8; transferred move d8e7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2436` (ply 23) [structural distance $d=102$, QAP obj=66]: the master played Nb6; under μ our Nd2 plays the role of that game's Nd2; cross-type roles: our Nb8 as their Ra8, our qg3 as their nf3; transferred move d7c5. Synthesis: 2 retrieved analogs transferred to Qe7 (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



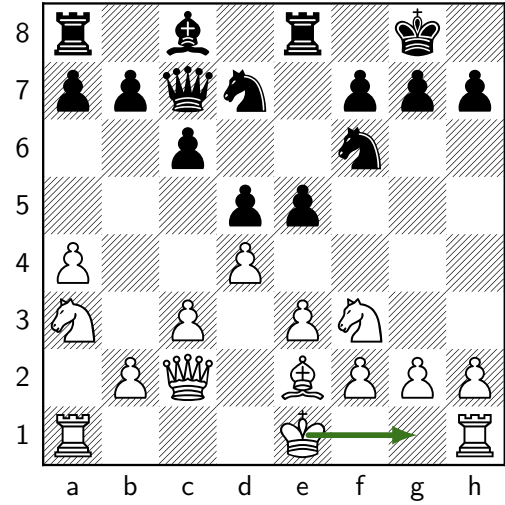
(a)



(b)

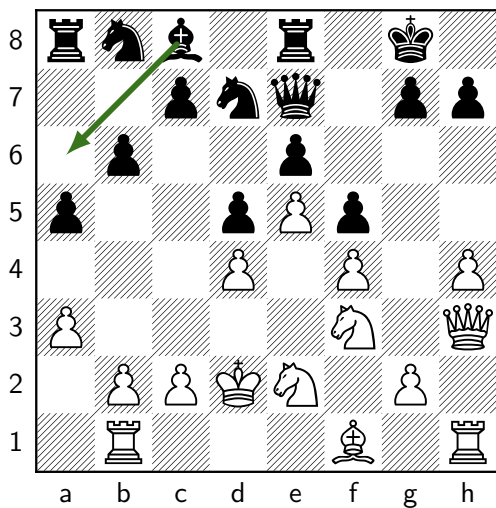


(c)

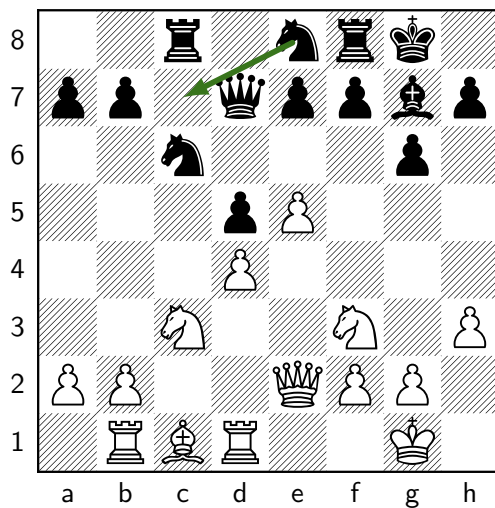


(d)

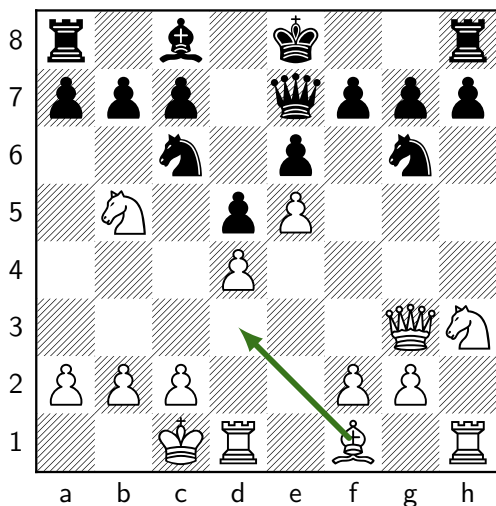
Figure 14: **Move 20, Black: Kg8**. (a) Our position; Melanie plays Kg8 (f7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3732` (ply 31) [structural distance $d=120$, QAP obj=52]: the master played 0-0; under μ our Kf2 plays the role of that game’s Ke1; cross-type roles: our Pa5 as their Ra8, our Nb8 as their Bc8; transferred move f7g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3997` (ply 23) [structural distance $d=108$, QAP obj=22]: the master played 0-0; under μ our Kf2 plays the role of that game’s Ke1; cross-type roles: our Pc7 as their Rh8, our rh1 as their Pb7; transferred move f7g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2492` (ply 24) [structural distance $d=88$, QAP obj=21]: the master played 0-0; under μ our Kf2 plays the role of that game’s Ke1; cross-type roles: our Qe7 as their Ra1, our Pc7 as their Rh1; transferred move f7g8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg8 (summed QAP objective 114, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



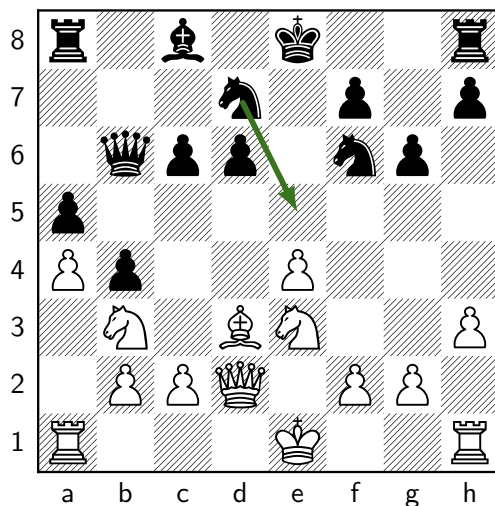
(a)



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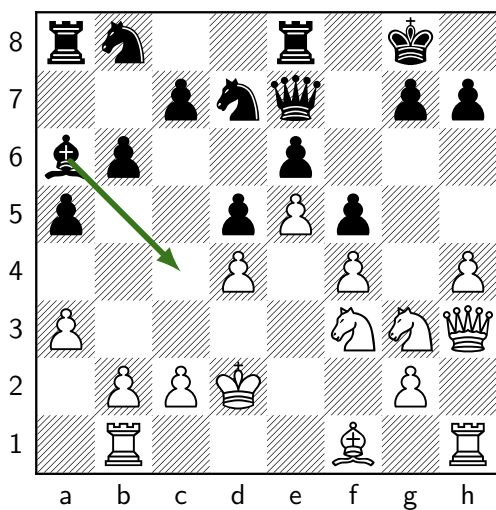


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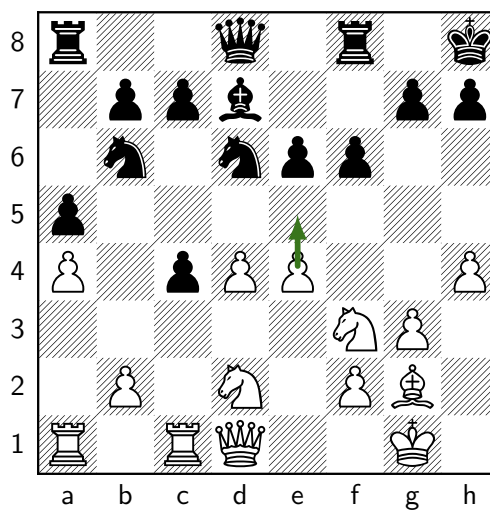


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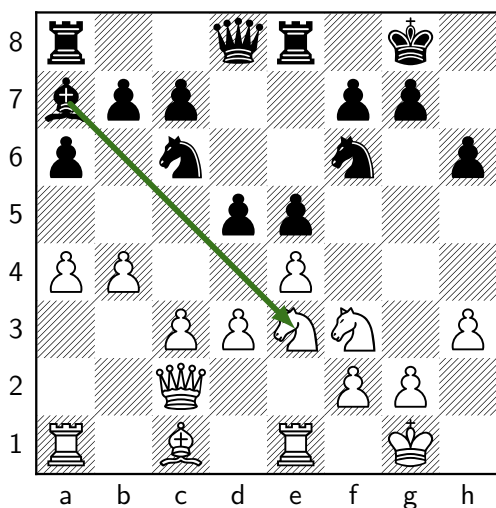
Figure 15: **Move 21, Black: Ba6**. (a) Our position; Melanie plays Ba6 (c8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1074` (ply 27) [structural distance $d=96$, QAP obj=92]: the master played Nc7; under μ our Bc1 plays the role of that game’s Ne1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Rc8, our Nd7 as their Bg7; transferred move c8a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#448` (ply 24) [structural distance $d=96$, QAP obj=39]: the master played Bd3; under μ our Bc1 plays the role of that game’s Bf1; cross-type roles: our Pe6 as their Rh1, our Nb8 as their Pb2; transferred move c8a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2891` (ply 31) [structural distance $d=86$, QAP obj=32]: the master played Ne5; under μ our Bc1 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Ra8, our Ph7 as their Bc8; transferred move c8a6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ba6 (summed QAP objective 164, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



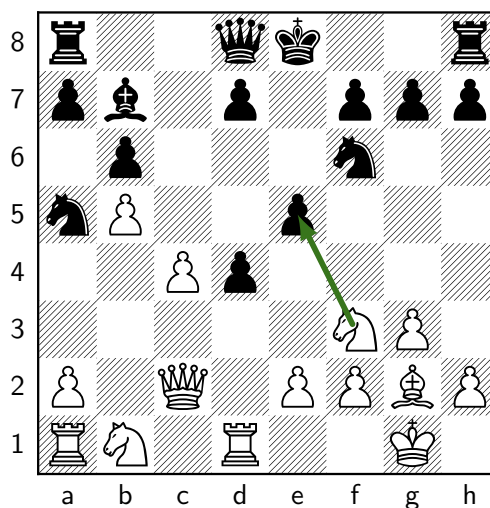
(a)



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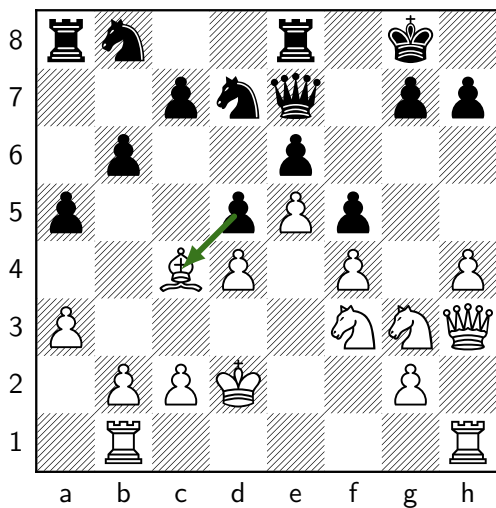


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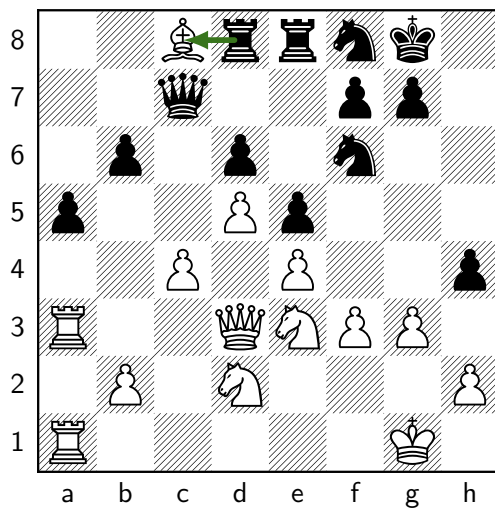


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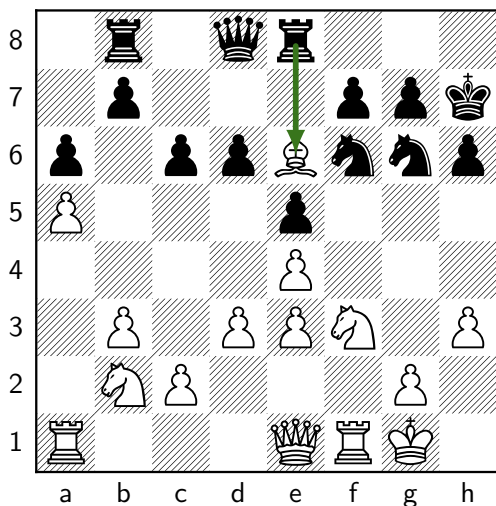
Figure 16: **Move 22, Black: Bc4**. (a) Our position; Melanie plays Bc4 (a6c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#509 (ply 30) [structural distance $d=98$, QAP obj=77]: the master played e5; under μ our *Ba3* plays the role of that game’s *Pe4* (a **cross-type** correspondence); cross-type roles: our *qh3* as their *Rc1*, our *pg2* as their *Qd1*; transferred move a6c4 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#547 (ply 27) [structural distance $d=106$, QAP obj=66]: the master played Bxe3; under μ our *Ba3* plays the role of that game’s *Ba2*; cross-type roles: our *Pb6* as their *Qd8*, our *Qe7* as their *Pb7*; transferred move a6c4 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#2859 (ply 26) [structural distance $d=94$, QAP obj=60]: the master played Nxe5; under μ our *Ba3* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Ba6* as their *Rd1*, our *Nb8* as their *Pe2*; transferred move a6c4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bc4 (summed QAP objective 204, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



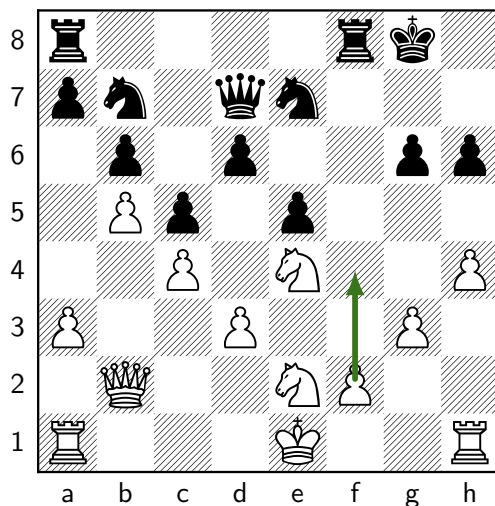
(a)



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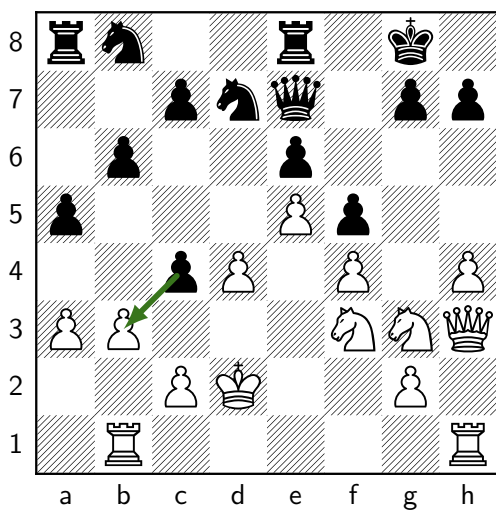


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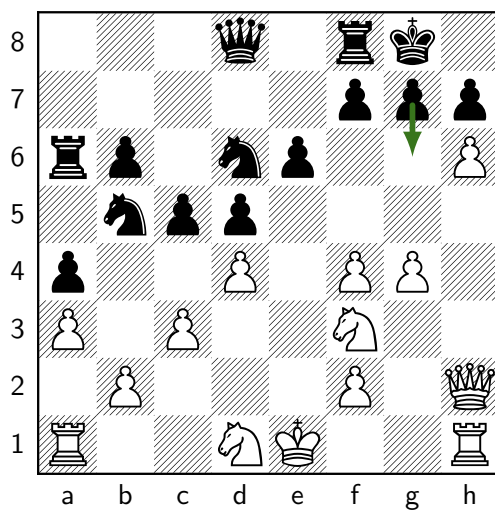


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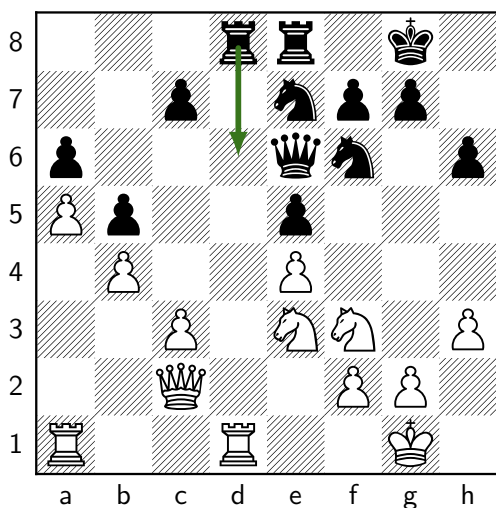
Figure 17: **Move 23, Black: dxc4.** (a) Our position; Melanie plays dxc4 (d5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1980` (ply 43) [structural distance $d=99$, QAP obj=39]: the master played Rxc8; under μ our *Pd4* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *Pe6* as their *Re8*, our *pg2* as their *Nf8*; transferred move d5c4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#467` (ply 35) [structural distance $d=93$, QAP obj=34]: the master played Rxe6; under μ our *Pd4* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *pc2* as their *Qd8*, our *Nb8* as their *Pc6*; transferred move d5c4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1390` (ply 36) [structural distance $d=92$, QAP obj=9]: the master played f4; under μ our *Pd4* plays the role of that game’s *Pf2*; cross-type roles: our *Pe6* as their *Ne4*, our *Nb8* as their *ph6*; transferred move d5c4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to dxc4 (summed QAP objective 110, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



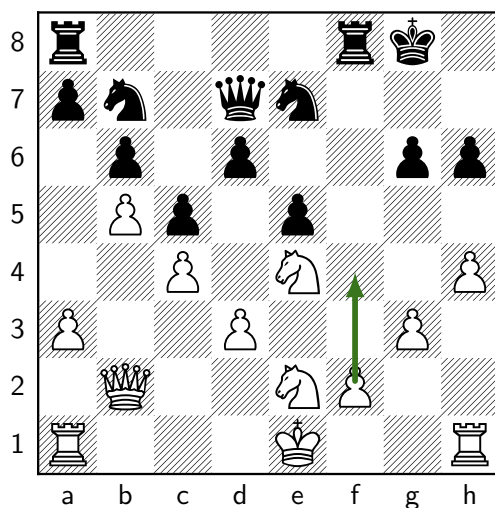
(a)



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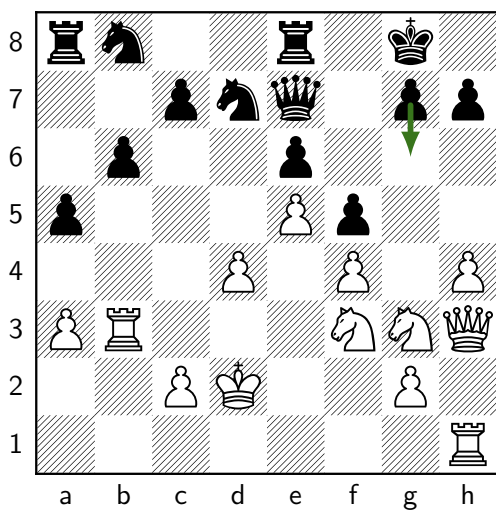


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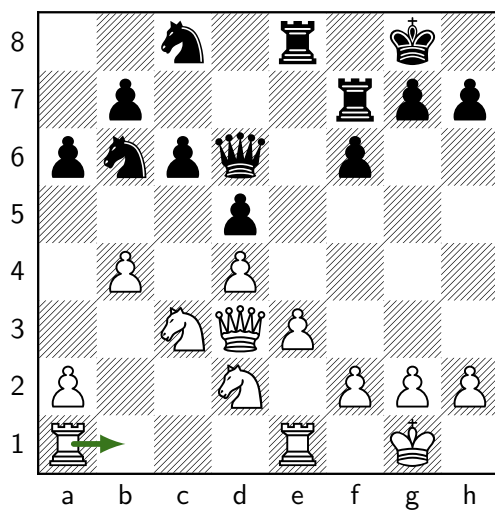


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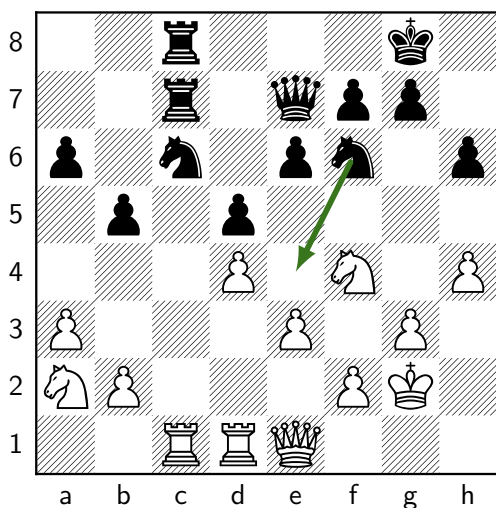
Figure 18: **Move 24, Black: cxb3**. (a) Our position; Melanie plays cxb3 (c4b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3345 (ply 39) [structural distance $d=85$, QAP obj=86]: the master played g6; under μ our Pc5 plays the role of that game’s Pg2; cross-type roles: our Nd7 as their Qd8, our pe5 as their Ra6; transferred move c4b3 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#8 (ply 47) [structural distance $d=89$, QAP obj=62]: the master played Rd6; under μ our Pc5 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Ne7, our Nb8 as their Pf7; transferred move c4b3 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#1390 (ply 36) [structural distance $d=82$, QAP obj=42]: the master played f4; under μ our Pc5 plays the role of that game’s Pf2; cross-type roles: our Pg7 as their Ne2, our Nb8 as their Pd3; transferred move c4b3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to cxb3 (summed QAP objective 214, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



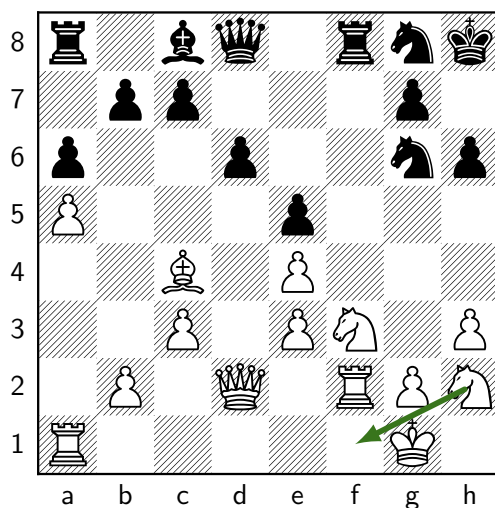
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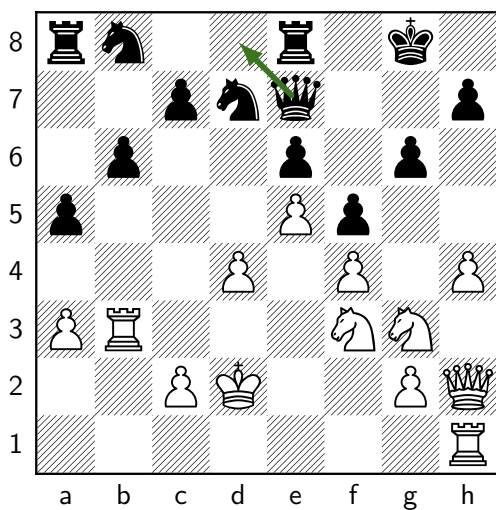


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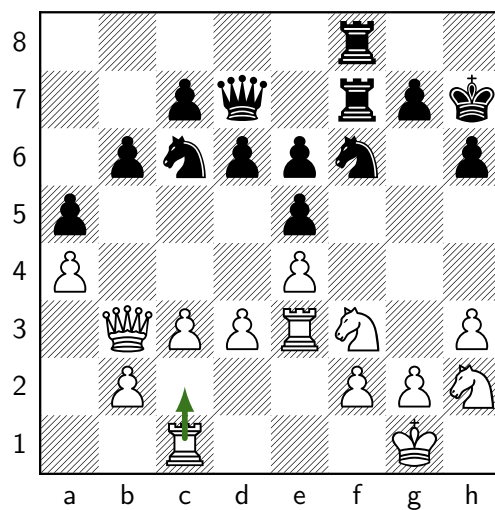


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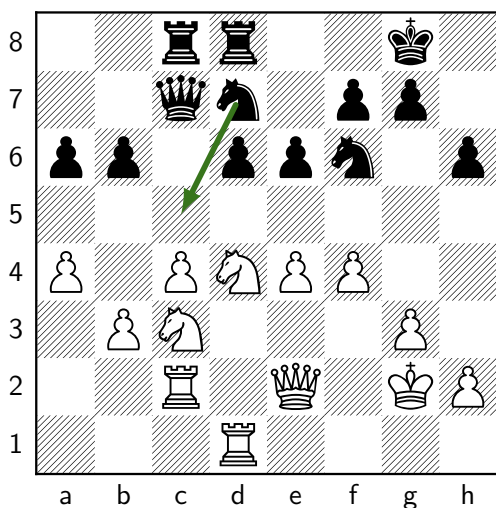
Figure 19: **Move 25, Black: g6**. (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1222` (ply 34) [structural distance $d=90$, QAP obj=97]: the master played Rab1; under μ our Pg2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Re1, our Re8 as their Pa2; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3484` (ply 55) [structural distance $d=91$, QAP obj=64]: the master played Ne4; under μ our Pg2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Rc7, our Nb8 as their Pa6; transferred move g7g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2560` (ply 36) [structural distance $d=96$, QAP obj=53]: the master played Nf1; under μ our Pg2 plays the role of that game’s Nh2 (a **cross-type** correspondence); cross-type roles: our rb3 as their Qd2, our Pa5 as their Rf2; transferred move g7g6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to g6 (summed QAP objective 230, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



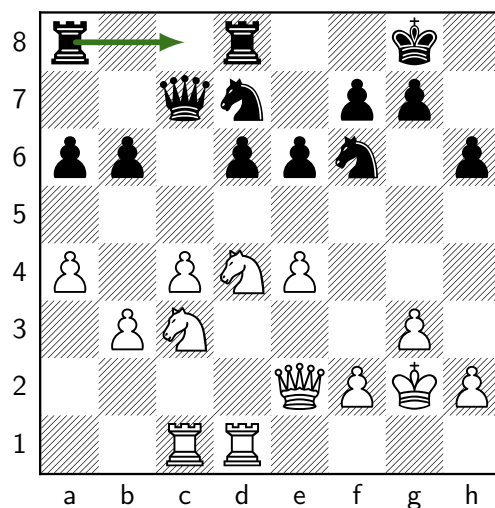
(a)



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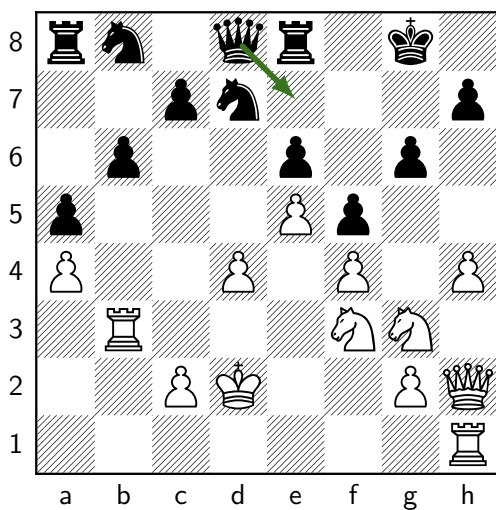


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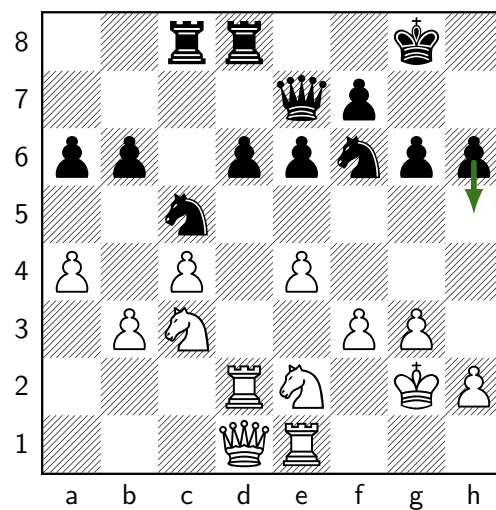


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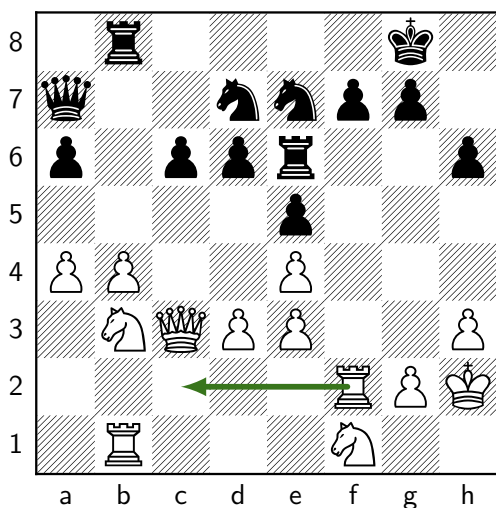
Figure 20: **Move 26, Black: Qd8**. (a) Our position; Melanie plays Qd8 (e7d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1163` (ply 34) [structural distance $d=99$, QAP obj=82]: the master played Rc2; under μ our Qe2 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Nh2, our rb3 as their Qb3; transferred move e7d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2016` (ply 41) [structural distance $d=87$, QAP obj=76]: the master played Nc5; under μ our Qe2 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Rc8, our Ph7 as their Rd8; transferred move e7d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#536` (ply 35) [structural distance $d=92$, QAP obj=75]: the master played Rac8; under μ our Qe2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our nf3 as their Rd8, our rb3 as their Qc7; transferred move e7d8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qd8 (summed QAP objective 315, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



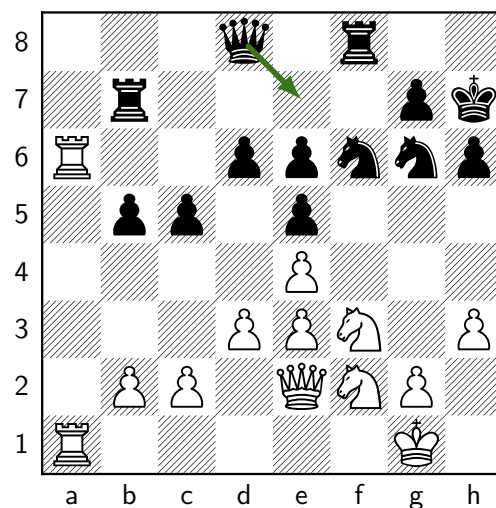
(a)



(b)



(c)



(d)

Figure 21: **Move 27, Black: Qe7**. (a) Our position; Melanie plays Qe7 (d8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#536` (ply 63) [structural distance $d=103$, QAP obj=99]: the master played h5; under μ our Qd1 plays the role of that game’s Ph3 (a **cross-type** correspondence); cross-type roles: our qh2 as their Rc8, our Nb8 as their Rd8; transferred move d8e7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3461` (ply 56) [structural distance $d=97$, QAP obj=81]: the master played Rc2; under μ our Qd1 plays the role of that game’s Rf2 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Rb1, our Re8 as their Nf1; transferred move d8e7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1937` (ply 45) [structural distance $d=75$, QAP obj=58]: the master played Qe7; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pc7 as their Rb7, our Nb8 as their Pg7; transferred move d8e7 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Qe7 (summed QAP objective 459, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

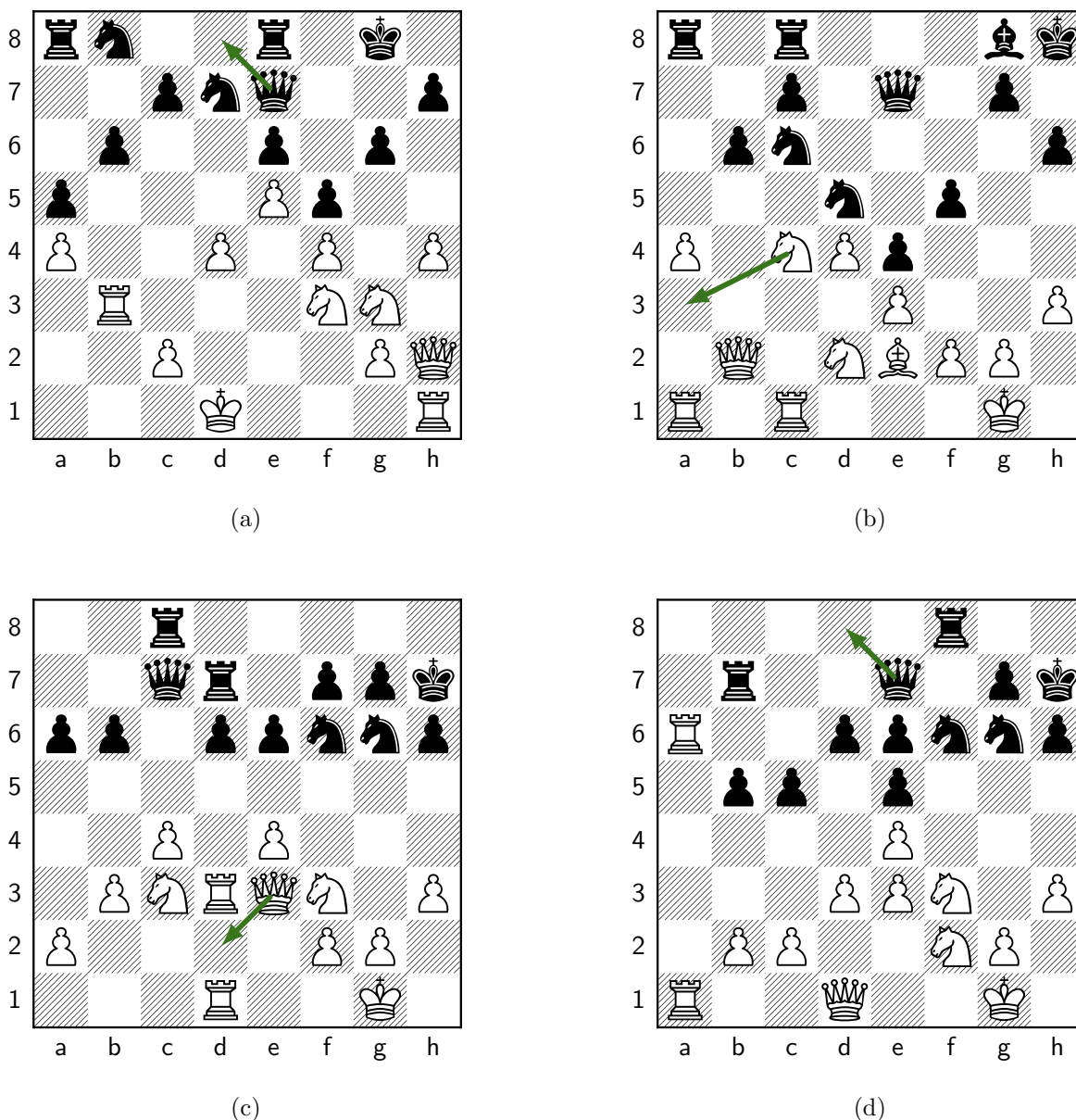
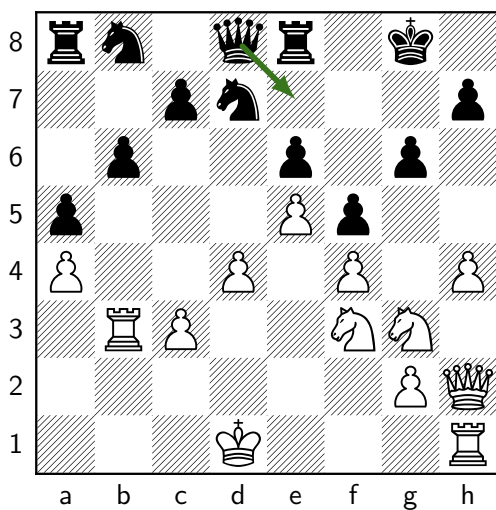
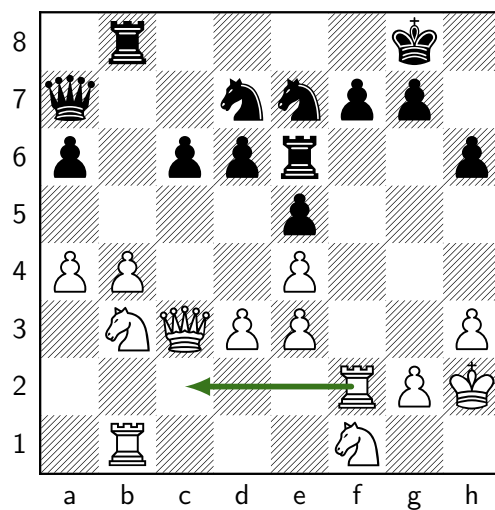


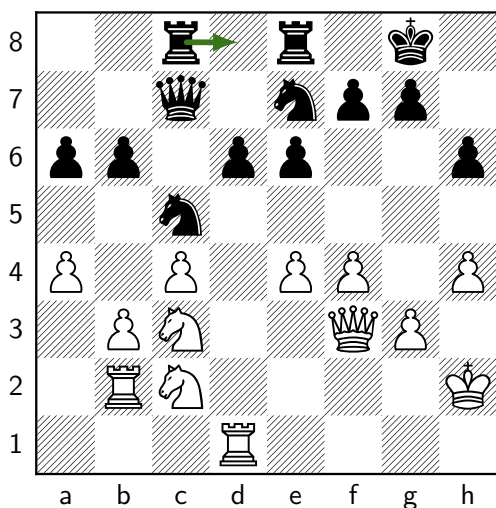
Figure 22: **Move 28, Black: Qd8.** (a) Our position; Melanie plays Qd8 (e7d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3596 (ply 48) [structural distance $d=103$, QAP obj=98]: the master played Na3; under μ our Qe2 plays the role of that game's Nc4 (a **cross-type** correspondence); cross-type roles: our Pa5 as their Ra1, our Pc7 as their Be2; transferred move e7d8 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#1330 (ply 60) [structural distance $d=95$, QAP obj=91]: the master played Qd2; under μ our Qe2 plays the role of that game's Qe3; cross-type roles: our Nb8 as their Pb3, our pc2 as their Nc3; transferred move e7d8 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#1937 (ply 43) [structural distance $d=82$, QAP obj=55]: the master played Qd8; under μ our Qe2 plays the role of that game's Qe2; cross-type roles: our pg2 as their Rf8, our Nb8 as their Pe6; transferred move e7d8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd8 (summed QAP objective 297, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



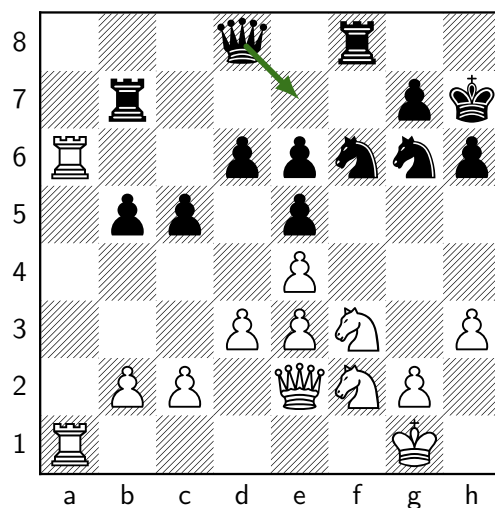
(a)



(b)

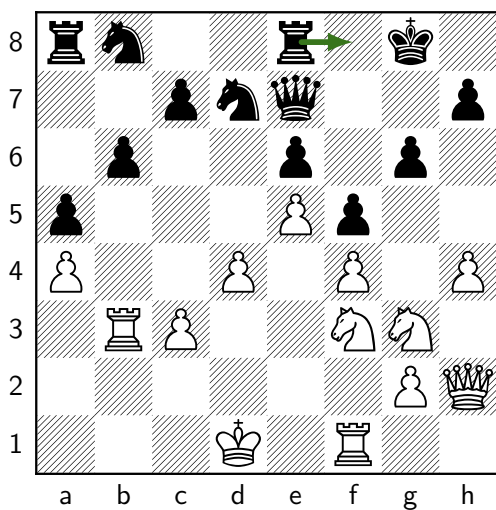


(c)

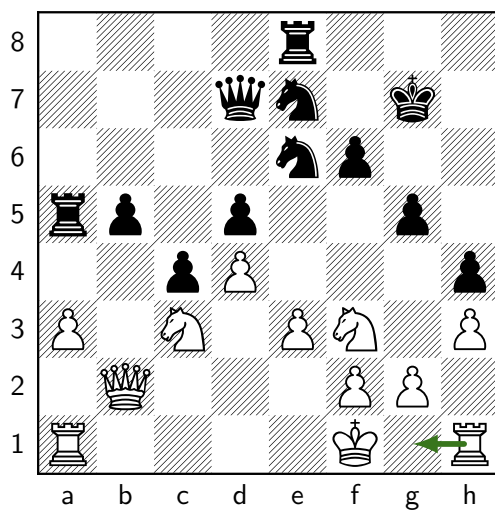


(d)

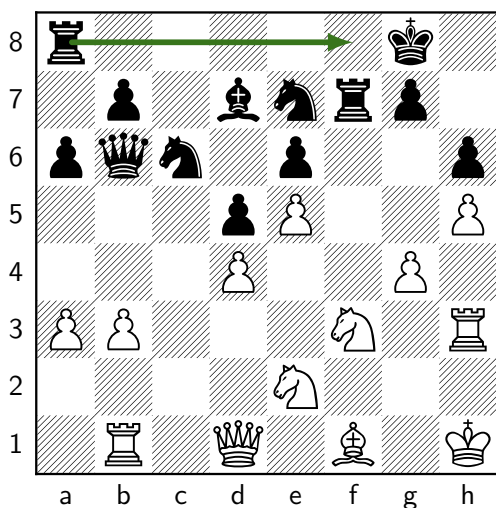
Figure 23: **Move 29, Black: Qe7.** (a) Our position; Melanie plays Qe7 (d8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3461` (ply 56) [structural distance $d=82$, QAP obj=86]: the master played Rc2; under μ our Qd1 plays the role of that game’s Rf2 (a **cross-type** correspondence); cross-type roles: our Pa5 as their Rb1, our Re8 as their Nf1; transferred move d8e7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#531` (ply 49) [structural distance $d=86$, QAP obj=70]: the master played Rcd8; under μ our Qd1 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our rb3 as their Ne7, our rh1 as their Pa6; transferred move d8e7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1937` (ply 45) [structural distance $d=74$, QAP obj=57]: the master played Qe7; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our ph4 as their Rf8, our Pc7 as their Rb7; transferred move d8e7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Qe7 (summed QAP objective 393, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



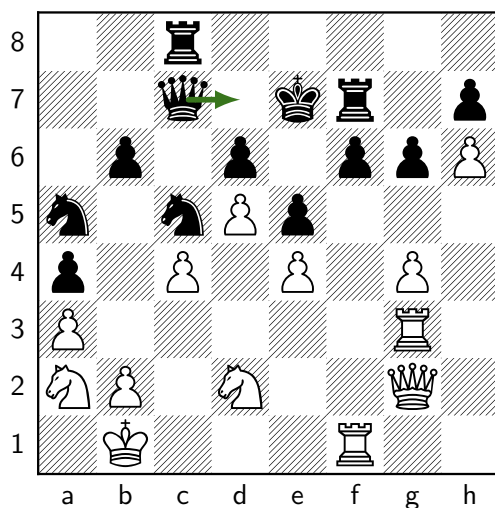
(a)



(b)

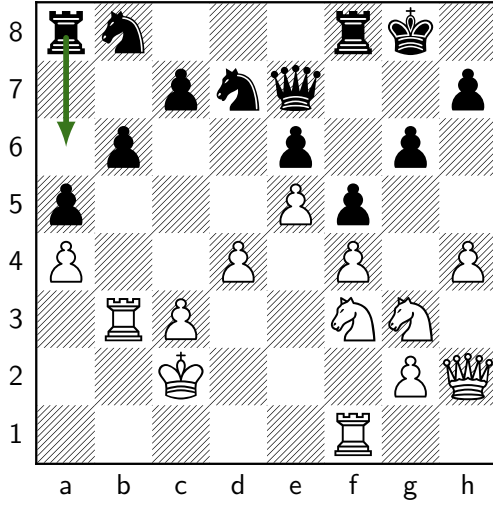


(c)

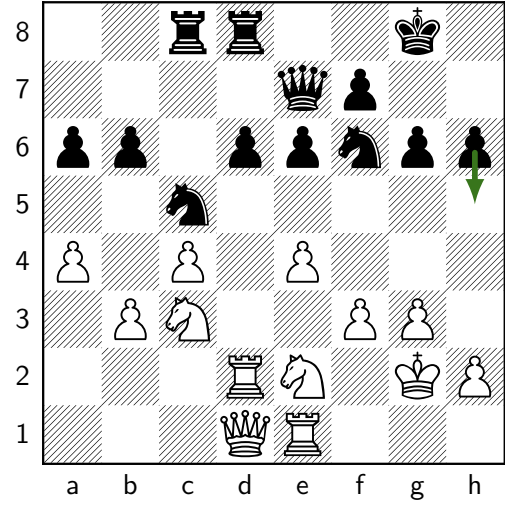


(d)

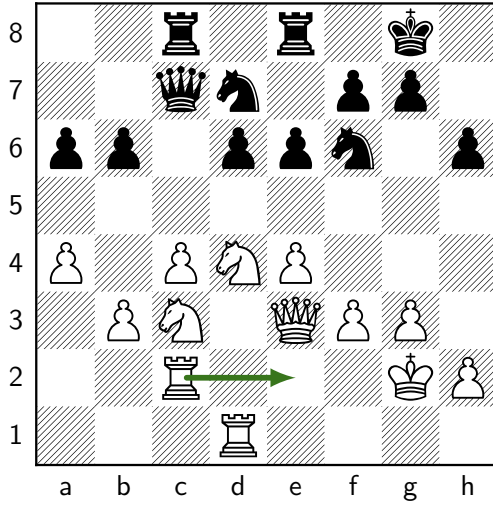
Figure 24: **Move 30, Black: Rf8 (e8f8).** (a) Our position; Melanie plays Rf8 (e8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3196` (ply 60) [structural distance $d=115$, QAP obj=123]: the master played Rg1; under μ our *Re1* plays the role of that game's *Rh1*; cross-type roles: our *Nb8* as their *Ra1*, our *Nd7* as their *Qb2*; transferred move e8f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3243` (ply 43) [structural distance $d=108$, QAP obj=71]: the master played Raf8; under μ our *Re1* plays the role of that game's *Ra1*; cross-type roles: our *qh2* as their *Pb7*, our *pg2* as their *Bd7*; transferred move e8f8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1706` (ply 63) [structural distance $d=91$, QAP obj=62]: the master played Qd7; under μ our *Re1* plays the role of that game's *Qc2* (a **cross-type** correspondence); cross-type roles: our *Qe7* as their *Rc8*, our *Nb8* as their *Pb6*; transferred move e8f8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rf8 (summed QAP objective 303, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



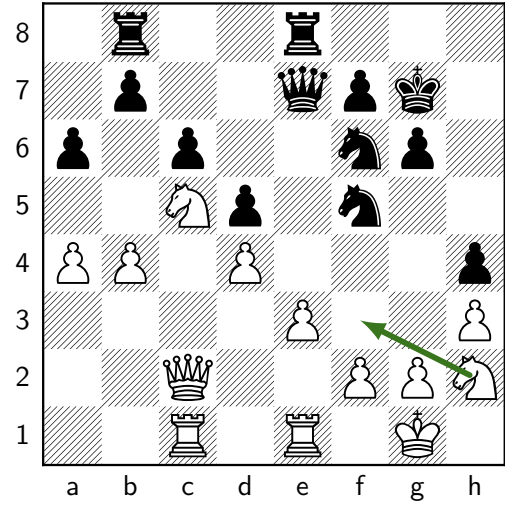
(a)



(b)

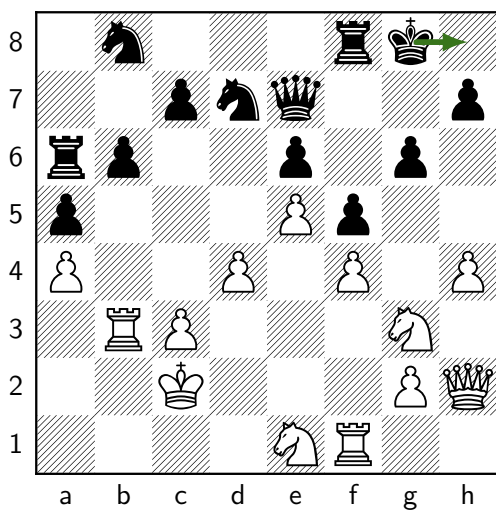


(c)

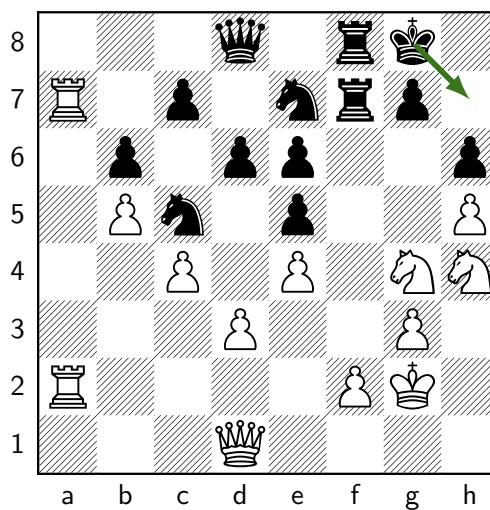


(d)

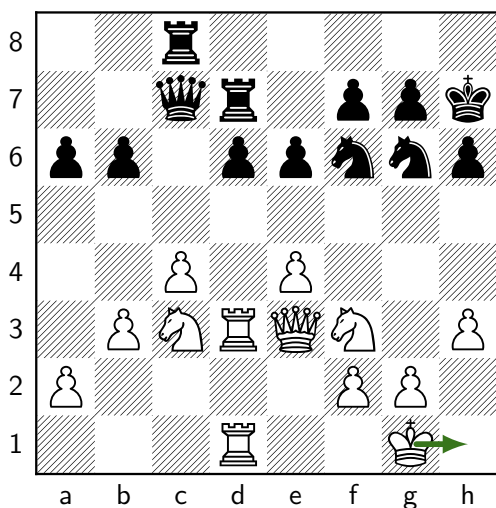
Figure 25: **Move 31, Black: Ra6**. (a) Our position; Melanie plays Ra6 (a8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#536` (ply 63) [structural distance $d=102$, QAP obj=97]: the master played h5; under μ our Ra1 plays the role of that game’s Ph3 (a **cross-type** correspondence); cross-type roles: our qh2 as their Rc8, our Nb8 as their Rd8; transferred move a8a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1595` (ply 46) [structural distance $d=105$, QAP obj=84]: the master played Re2; under μ our Ra1 plays the role of that game’s Rc2; cross-type roles: our pg2 as their Rd1, our Nd7 as their Qe3; transferred move a8a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3382` (ply 46) [structural distance $d=106$, QAP obj=36]: the master played Nf3; under μ our Ra1 plays the role of that game’s Nh2 (a **cross-type** correspondence); cross-type roles: our Qe7 as their Rc1, our nf3 as their Re1; transferred move a8a6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ra6 (summed QAP objective 218, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



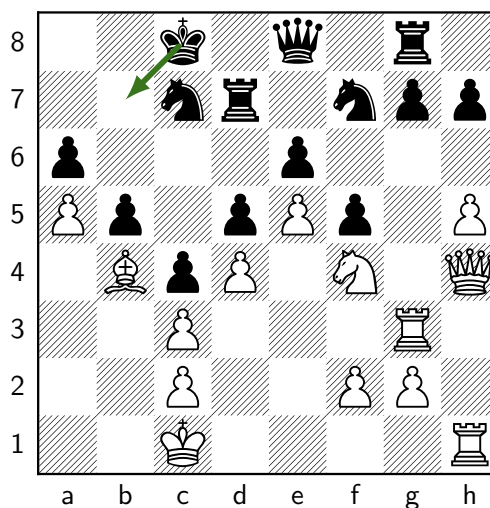
(a)



(b)

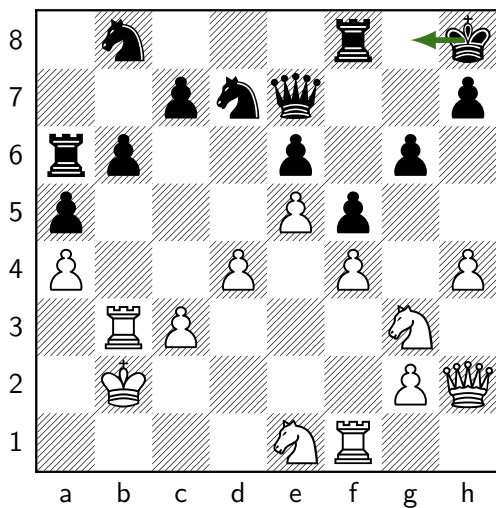


(c)

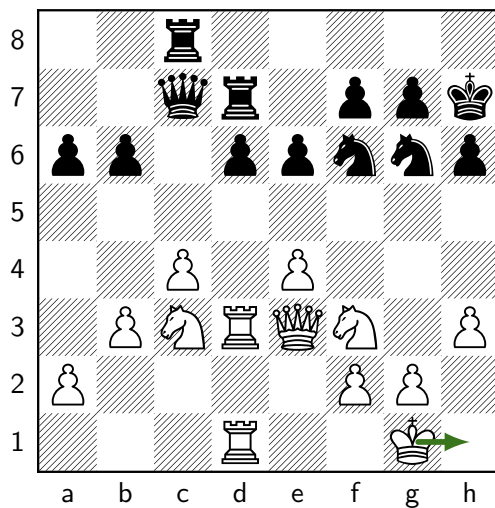


(d)

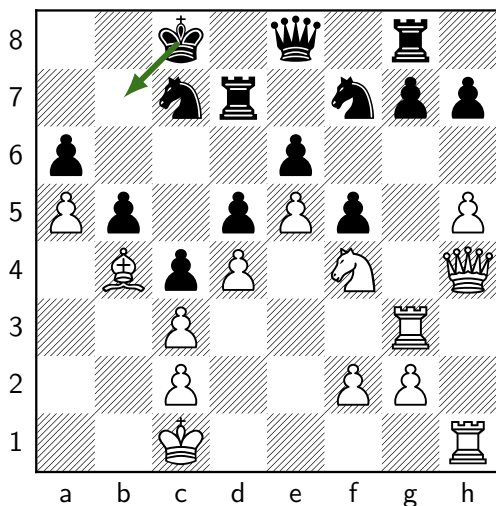
Figure 26: **Move 32, Black: Kh8**. (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3421` (ply 79) [structural distance $d=88$, QAP obj=116]: the master played Kh7; under μ our *Kg1* plays the role of that game's *Kg1*; cross-type roles: our *Ra6* as their *Qd8*, our *Qe7* as their *Rf7*; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1330` (ply 56) [structural distance $d=88$, QAP obj=89]: the master played Kh1; under μ our *Kg1* plays the role of that game's *Kg1*; cross-type roles: our *Ph7* as their *Rd1*, our *Ra6* as their *Pb3*; transferred move g8h8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#518` (ply 71) [structural distance $d=93$, QAP obj=79]: the master played Kb7; under μ our *Kg1* plays the role of that game's *Kc1*; cross-type roles: our *rf1* as their *Qe8*, our *Nb8* as their *Rg8*; transferred move g8h8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kh8 (summed QAP objective 468, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



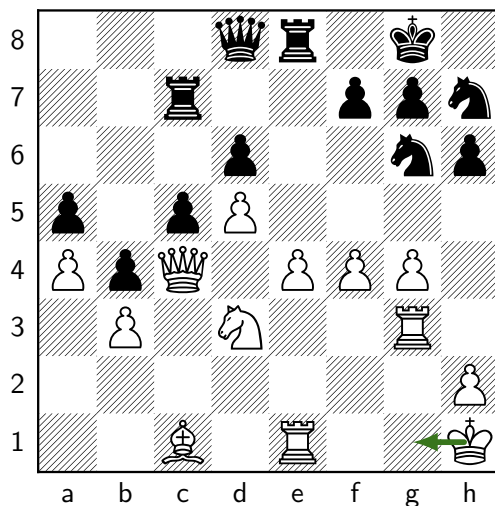
(a)



(b)

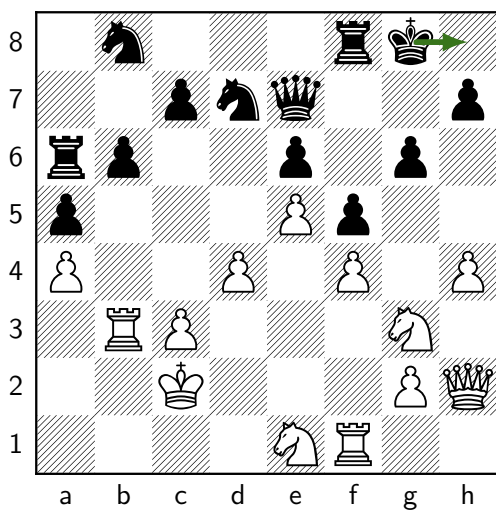


(c)

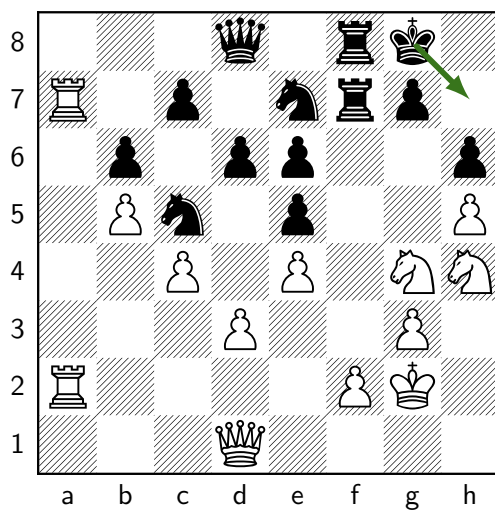


(d)

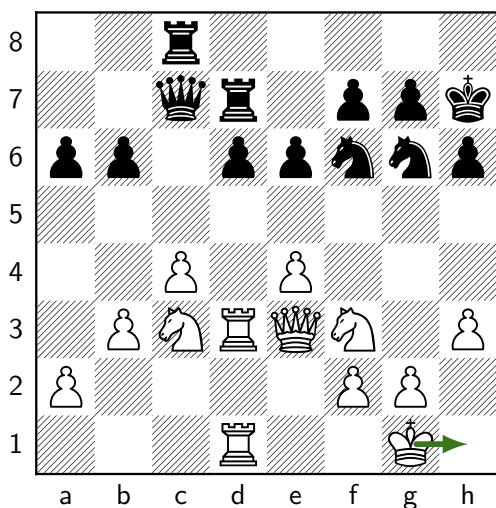
Figure 27: **Move 33, Black: Kg8.** (a) Our position; Melanie plays Kg8 (**h8g8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1330` (ply 56) [structural distance $d=89$, QAP obj=104]: the master played Kh1; under μ our Kh1 plays the role of that game’s Kg1; cross-type roles: our Nb8 as their Pa2, our Nd7 as their Pf2; transferred move **h8g8 (endorses this move)**. (c) Retrieved from `ccr1_top_p0003#518` (ply 71) [structural distance $d=88$, QAP obj=79]: the master played Kb7; under μ our Kh1 plays the role of that game’s Kc1; cross-type roles: our rf1 as their Qe8, our Nb8 as their Rg8; transferred move **h8g8 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#346` (ply 66) [structural distance $d=93$, QAP obj=60]: the master played Kg1; under μ our Kh1 plays the role of that game’s Kh1; cross-type roles: our Rf8 as their Bc1, our Nb8 as their Re1; transferred move **h8g8 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to Kg8 (summed QAP objective 288, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



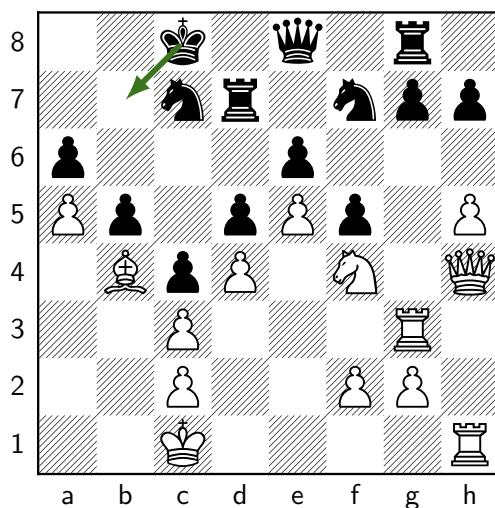
(a)



(b)

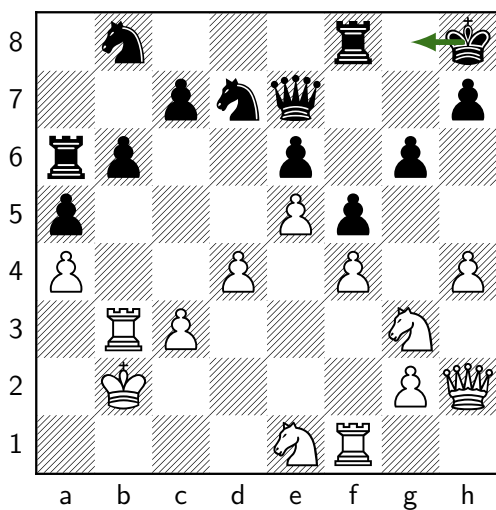


(c)

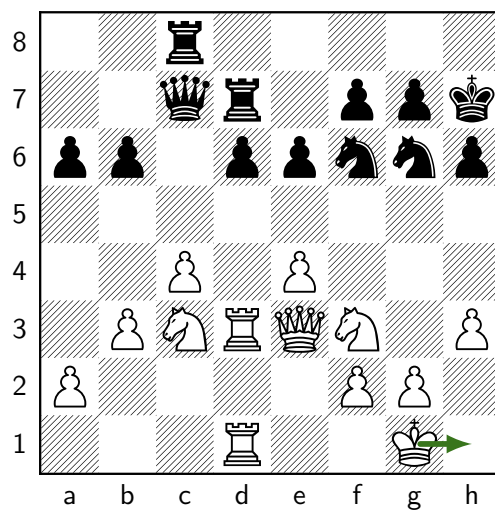


(d)

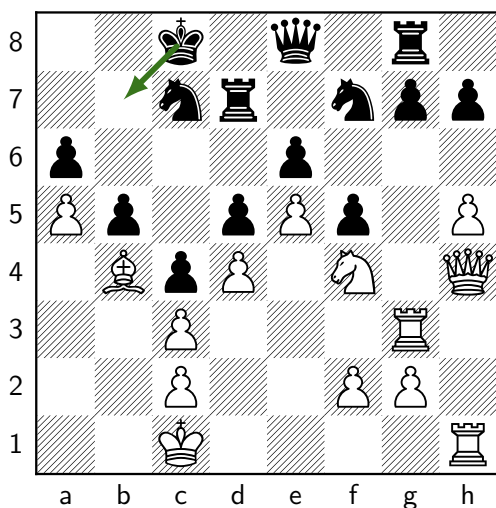
Figure 28: **Move 34, Black: Kh8 (g8h8).** (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3421` (ply 79) [structural distance $d=88$, QAP obj=116]: the master played Kh7; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Ra6* as their *Qd8*, our *Qe7* as their *Rf7*; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1330` (ply 56) [structural distance $d=88$, QAP obj=89]: the master played Kh1; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Ph7* as their *Rd1*, our *Ra6* as their *Pb3*; transferred move g8h8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#518` (ply 71) [structural distance $d=93$, QAP obj=79]: the master played Kb7; under μ our *Kg1* plays the role of that game’s *Kc1*; cross-type roles: our *rf1* as their *Qe8*, our *Nb8* as their *Rg8*; transferred move g8h8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kh8 (summed QAP objective 469, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



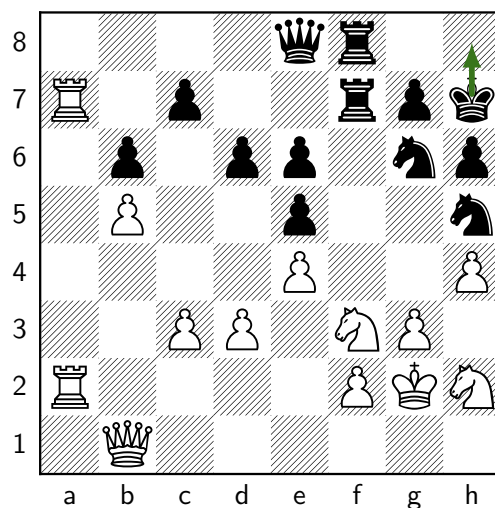
(a)



(b)



(c)



(d)

Figure 29: **Move 35, Black: Kg8 (h8g8).** (a) Our position; Melanie plays Kg8 (h8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1330` (ply 56) [structural distance $d=89$, QAP obj=104]: the master played Kh1; under μ our Kh1 plays the role of that game's Kg1; cross-type roles: our Nb8 as their Pa2, our Nd7 as their Pf2; transferred move h8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#518` (ply 71) [structural distance $d=88$, QAP obj=79]: the master played Kb7; under μ our Kh1 plays the role of that game's Kc1; cross-type roles: our rf1 as their Qe8, our Nb8 as their Rg8; transferred move h8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3421` (ply 65) [structural distance $d=96$, QAP obj=62]: the master played Kh8; under μ our Kh1 plays the role of that game's Kh2; cross-type roles: our Nd7 as their Rf7, our Qe7 as their Ng6; transferred move h8g8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kg8 (summed QAP objective 351, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.